

Which Exchange Rates Matter to Firms? Trade Networks and Equity Currency Exposure*

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Stock prices rise with local-currency appreciation in some countries and fall in others. Standard trade theory accommodates both, since appreciation lowers both foreign revenue and imported input costs. Empirical work has struggled to detect either channel, because a common benchmark nets the two exposures against each other. This paper separates them, weighting currency returns by export destinations and by import origins. Appreciation against export destinations lowers stock prices while appreciation against import origins raises them. Revenue and input costs move as the theory predicts, with dollar invoicing delaying pass-through. Which channel dominates depends on gross profit margin. A higher margin implies less weight on imported-input savings, so higher-margin economies have more negative dollar betas. The relationship is robust to controls for foreign-currency balance-sheet exposures.

JEL classification: G15, F41, F14

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Abstract

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1 Introduction

When a country's currency appreciates, its exporters lose competitiveness abroad while its producers pay less for imported inputs. The first effect reduces revenue and the second costs, so the two move profits and equity values in opposite directions. Classical treatments of exchange rate exposure recognize both (Shapiro, 1975; Dumas, 1978). Empirical work, however, usually measures exposure as the slope of equity returns on a single exchange rate series, usually the dollar rate. What is seen in the data is a single coefficient, which is the net of two offsetting forces. The sign of the coefficient indicates which force dominates, not how large either one is.

Figure 1 plots, for each country, the estimated USD currency betas of country index returns in excess of the U.S. market. Hau and Rey (2006), for example, use the exact regression form for 17 OECD markets and find strong negative comovement. In a broader sample, the figure shows that the betas differ across countries not only in magnitude but in sign.¹ Betas tend to be positive in emerging markets and negative in developed markets, with China, Thailand, and Taiwan at one end of the distribution and Israel, the United Kingdom, and Japan at the other. The sign is not a matter of development alone, however. Betas are positive in Norway and Singapore and negative in Malaysia and South Africa. Whatever the USD beta captures, it is something that is not the same in Thailand as in Japan.

The empirical literature has struggled to detect the relationship of this to the trade channel. Early evidence, largely from the United States, finds little detectable exchange rate

¹That the sign could differ across countries is unsurprising. For example, Pavlova and Rigobon (2007) show that in a general equilibrium framework, real exchange rates can comove with stocks in either direction depending on whether demand or output shocks dominate.

exposure (Jorion, 1991; Bartov and Bodnar, 1994). Later work documents that exposure varies over time and across countries (Dominguez and Tesar, 2006; Cenedese, Payne, Sarno, and Valente, 2016), that it is typically negative in developed markets (Hau and Rey, 2006) and positive in emerging markets (Cho, Choi, Kim, and Kim, 2016), and that firms hedge their own gross exposure through price pass-through (Bartram, Brown, and Minton, 2010; Amiti, Itskhoki, and Konings, 2014). The trade channel may be obscured in the data because firms hedge gross exposure (Bartram, Brown, and Minton, 2010), or because two-way trade naturally offsets export and import exposures (Amiti, Itskhoki, and Konings, 2014). Trade does enter these studies, but as a control or industry attribute such as export share, import share, or the markup (Allayannis and Ihrig, 2001; Dominguez and Tesar, 2006). In these studies, the export and import channels remain inside the single benchmark currency, and their separate roles have to be inferred from an interaction rather than observed directly.

The trade network provides the missing distinction. Conventional trade-weighted exchange rates pool export and import partners into one index, obscuring the fact that currency movements against customers and suppliers often diverge and have opposing effects. A currency appreciation against markets an industry sells to should lower stock returns, whereas an appreciation against markets from which it buys inputs from should raise them. This paper separates these two channels into export-destination-weighted and import-origin-weighted currency returns. The empirical analysis tests these predictions at the country-industry level using input-output data, and at the country level using two independent measures of the trade network.

Which force dominates depends not only on the trade network but also on the profitability of firms. The trade network determines which currencies matter for sales and which for input costs. Hence, it determines against whom an appreciation hurts and against whom it helps.

The gross profit margin determines the relative gain from cheaper inputs it receives. A low-margin economy spends a larger share of revenue on inputs, so it gains more from cheaper imports, and its dollar beta can be positive. A high-margin economy has a smaller cost to save and the loss from revenue is relatively greater, so its beta is more negative. This explains why the same dollar appreciation is associated with lower equity values in Japan and higher values in Thailand.

At the country-industry level, the empirical results show that export-relative appreciations are associated with lower monthly returns and import-relative appreciations with higher returns. The export-to-import differential has a positive effect on stock returns controlling for USD returns, while the sign of the USD coefficient varies across specifications and is mostly statistically insignificant. The differential increases with greater international trade exposure. Within countries, it is larger for tradable than for non-tradable industries and for goods than for service industries. Across countries, it is larger in more open economies.

The accounting data provide direct evidence of the proposed channels. Appreciations relative to export partners are associated with lower revenue, and appreciations relative to import origins are associated with lower cost of goods sold. The estimated responses are weaker contemporaneously but become clearer at lagged and cumulative horizons. Higher USD invoicing shares are associated with weaker contemporaneous revenue and cost responses, while this attenuation is only strong in the same year and becomes statistically insignificant at longer horizons.

At the country level, both measures using the input-output and customs-based data yield export and import coefficients with the predicted signs. The export-to-import differential is significant, whereas the bilateral USD return is not. A one-standard-deviation export-to-

import shock is associated with a 26-basis-point monthly country-index return. Net trade measures do not always significantly affect the currency beta after aggregating to country level, which is possibly one reason previous literature struggles to identify a trade channel.

Gross profit margins are negatively related to USD currency betas in the baseline country specifications and remain after controlling for valuation, net-trade, net international investment position, and foreign-currency balance-sheet exposures. In the cross section, the correlation between profit margins and USD currency betas is -0.55 .

The relationship between gross profit margin and USD currency beta is mostly a variation across countries and across country industries, but also varies across time within country and industry. When a country's profit margin rises relative to its own average, its USD currency beta becomes more negative. The same pattern holds at the country-industry level. Moreover, the margin effect is stronger in more trade-intensive economies, consistent with the proposed export-revenue and import-cost channels.

Finally, I construct a currency beta implied entirely by trade-based revenue and cost elasticities, without using any covariation between stock and currency returns, and ask how well it tracks the realized equity beta estimated from return covariances. Across countries, the two are positively related, with a correlation between 0.44 and 0.54 depending on the aggregation weights. While a substantial fraction of variation still remains unexplained, this accounting-based comparison indicates that trade-based revenue and cost channels account for a non-negligible share of cross-country variation in equity currency exposure, without relying on return covariances.

The paper makes three contributions. First, the bilateral export- and import-weighted currency returns separate the opposing trade channels, which in previous literature is often

obscured by a common currency benchmark. Second, it relates the return patterns to firm operation data, such as revenue, costs, profit margins. Third, it documents the same patterns at the country level and shows that gross profit margins help explain the sign of aggregate USD currency betas, with additional evidence from variation within country-industries over time.

This paper is most directly related to work on the goods-market channel of equity currency exposure, where the evidence is mixed. Bodnar, Dumas, and Marston (2002) show that exchange-rate exposure and pass-through depend on product-market conditions, while Allayannis and Ihrig (2001) find opposing export- and import-side effects of dollar appreciation on industry returns. This paper separates currency shocks relative to export destinations and import origins. Its profitability mechanism is also distinct from the markup mechanism in Allayannis and Ihrig (2001). In this paper, a higher gross profit margin reduces the cost share and therefore the weight on imported-input savings, making the export-revenue channel relatively more important for the USD beta.²

A related literature examines how exchange rate movements affect firms' accounting values. Welch and Zhou (2024) and Adams and Verdelhan (2025) show that limited hedging leaves revenues, costs, and profits sensitive to currency fluctuations, while Amiti, Itskhoki, and Konings (2014) and Gopinath and Rigobon (2008) emphasize incomplete pass-through and sticky invoicing-currency prices. Consistent with these papers, the revenue and cost responses documented below appear with a lag, and a higher USD invoicing share delays rather than eliminates them.

²Related work documents time variation in currency exposure, limited evidence for simple industry trade measures, and the role of operational and financial hedging (Dominguez and Tesar, 2006; Bartram, Brown, and Minton, 2010). See also Bodnar, Simon, and Gentry (1993) and Griffin and Stulz (2001) for additional evidence on goods-market currency exposure.

A significant fraction of the equity currency exposure is related to the goods channel, but this is obviously not the only channel. Hau and Rey (2006) document a negative relationship for developed economies but attribute it to portfolio rebalancing rather than trade. This channel is formalized in equilibrium by Camanho, Hau, and Rey (2022) and extended to institutional hedging flows by Ben Zeev and Nathan (2024), among others. Related work also emphasizes the role of external balance sheets and currency mismatch at the country level (e.g., Gourinchas and Rey, 2007; Lane and Shambaugh, 2010; Bénétrix, Lane, and Shambaugh, 2015) and firm-level foreign-currency debt (Kim, Tesar, and Zhang, 2015; Bruno and Shin, 2017; Kalemli-Ozcan, Liu, and Shim, 2021). Relatedly, Colacito, Qian, and Stathopoulos (2025) show that firms align the currency composition of debt with the geographic distribution of sales, consistent with financial hedging of operating exposure. These channels are not mutually exclusive with the mechanism of this paper and may account for variation left unexplained by trade exposure. However, if foreign-currency debt is denominated mainly in dollars and euros, its balance-sheet effect should load on the bilateral USD return or EUR return rather than on the export-to-import differential.

Finally, this paper builds on work that uses bilateral trade networks to explain asset prices. One strand examines equity-market transmission. Forbes (2004) shows that firms competing in export markets with crisis countries suffered lower returns during the Asian and Russian crises. Albuquerque, Ramadorai, and Watugala (2015) find that returns in customer countries predict future returns of producer firms. Huang, Lin, Liu, and Tang (2023) study the propagation of U.S.–China tariff shocks through the production network. Pyun and Sulaeman (2026) show that stock returns of countries that compete more intensely in common export markets have lower stock-return comovement. A second strand relates trade networks to currency risk premia and exchange-rate risk (Richmond, 2019; Hassan,

Loualiche, Pecora, and Ward, 2023; Hou, Sarno, and Ye, 2025). This paper instead treats exchange-rate movements as inputs and studies their effects on stock returns through export-side and import-side networks.

The remainder of this paper is organized as follows. Section 2 presents a simple framework that illustrates the intuition. Section 3 explains the data and construction of currency returns. Section 4 provides the country-industry level result, followed by the country-level analysis in Section 5. The paper concludes in Section 6.

2 The framework

Exchange-rate movements can affect firm profits through two opposing channels. A local currency appreciation could reduce foreign-sales revenue, but it could also lower the cost of imported inputs. Because firms sell to and buy from different countries, the exchange rates relevant to the two channels need not be the same. This distinction raises two questions. Which bilateral exchange rates affect operating profit? How do the resulting revenue and cost responses combine in an equity beta measured against a single currency? This section develops a framework to address these two questions. Appendix A provides microfoundations for the framework, derives the responses in closed form, and shows how dominant-currency invoicing affects short-run exposure.

2.1 Bilateral exchange rates and operating profit

Consider a firm with foreign trading partners indexed by d , and let q_d be the value of the firm's home currency against partner d 's currency. A rise in q_d is a home-currency

appreciation. Let Π^* denote operating profit evaluated at the firm's optimal choices, S_d^X its export revenue from partner d , and C_d^M its spending on inputs imported from partner d , all measured in the home currency.

If the exchange rate affects profit directly only by converting foreign-currency sales and input spending into the home currency, Appendix A shows that optimized profit responds to a change in the bilateral rate by:

$$\frac{\partial \Pi^*}{\partial \ln q_d} = -S_d^X + C_d^M. \quad (1)$$

An appreciation against a customer lowers the home-currency value of export revenue, whereas an appreciation against a supplier lowers imported-input cost.

Summing across partners:

$$d\Pi^* = -S^X d\ln q^X + C^M d\ln q^M, \quad (2)$$

where $S^X = \sum_d S_d^X$ and $C^M = \sum_d C_d^M$, and

$$d\ln q^X \equiv \sum_d \frac{S_d^X}{S^X} d\ln q_d, \quad d\ln q^M \equiv \sum_d \frac{C_d^M}{C^M} d\ln q_d. \quad (3)$$

$d\ln q^X$ and $d\ln q^M$ are two exposure-weighted summaries of the same vector of bilateral currency movements, using customer weights for the revenue channel and supplier weights for the cost channel. The result aggregates across firms and therefore also applies at the country-industry and country levels.

Assume that changes in operating profit pass directly one-to-one into dividends, so that equity value is the present value of expected operating profit. If expected profit grows at a constant rate g and is discounted at δ , with $\gamma \equiv \delta(1 + g) < 1$, the stock price is determined by:

$$P_t = \sum_{s=0}^{\infty} \delta^s E_t \Pi_{t+s} = \Lambda \Pi_t, \quad \Lambda = \frac{1}{1 - \gamma}. \quad (4)$$

A persistent profit shock moves equity value by $\Lambda d\Pi_t$, so Equation (2) gives

$$\frac{dP_t}{P_t} \approx \Lambda \frac{S_t}{P_t} \left[-\theta^X d \ln q_t^X + \frac{C_t}{S_t} \theta^M d \ln q_t^M \right], \quad (5)$$

where $\theta^X = S^X/S$ is the share of revenue earned abroad and $\theta^M = C^M/C$ the share of cost paid to foreign suppliers. The stock return is negatively related to the export-weighted currency return and positively related to the import-weighted currency return, with the magnitude of each loading increasing in the corresponding foreign exposure.

2.2 The cash-flow component of the currency beta

Existing literature measures currency exposure as a beta against one exchange rate, often the US dollar. To see what such a single currency beta contains, write profit as $\Pi_t = S_t - C_t + K_t$, where S_t is revenue, C_t is cost, and K_t collects other non-operating items, such as investment income/loss, interest, and taxes. Equity value is the present value of expected cash flows, and a currency shock can move it by changing expected cash flows or by changing the discount rate (Campbell and Shiller, 1988; Campbell, 1991). Consider fixed

discount weights δ^h and horizons $h = 0, \dots, H$ and define the discounted sum of responses of firm revenue and cost to returns on a currency (q_t) as

$$V_{q,H}^S \equiv \frac{1}{S_t} \sum_{h=0}^H \delta^h \frac{\partial E_t S_{t+h}}{\partial \ln q_t} \quad V_{q,H}^C \equiv \frac{1}{C_t} \sum_{h=0}^H \delta^h \frac{\partial E_t C_{t+h}}{\partial \ln q_t}, \quad (6)$$

respectively. Then, the contribution of revenue and cost to the proportional equity-value response is

$$\beta_{q,H}^{CF} \equiv \frac{1}{P_t} \sum_{h=0}^H \delta^h \left(\frac{\partial E_t S_{t+h}}{\partial \ln q_t} - \frac{\partial E_t C_{t+h}}{\partial \ln q_t} \right) = \frac{S_t}{P_t} \left(V_{q,H}^S - \frac{C_t}{S_t} V_{q,H}^C \right), \quad (7)$$

This identity uses no demand curve, market structure, or pricing regime, and because revenue, cost, and equity value add across firms, it applies unchanged at the firm, country-industry, and country levels. Every term in this equation is measurable, although it may contain noise. The sales-to-price ratio and the cost share are observed levels, and V^S and V^C can be estimated from the cumulative responses of revenue and cost to currency returns.

The decomposition is not designed to completely capture the variation in the realized regression beta. Responses of accounting variables outside the gross profit margin, which includes interest on foreign-currency debt, discount-rate news, financial hedging, and the portfolio-rebalancing channel of Hau and Rey (2006) and Camanho, Hau, and Rey (2022) can all move equity returns and are not in β^{CF} . A finite horizon H truncates a persistent response, so the cash flow currency beta omits responses beyond H years. Also, the decomposition assumes constant discount rates, so time-varying risk premia and the joint effect of the stochastic discount factor on stock and currency pricing are not captured by the decomposition. The comparison of implied and realized betas in a later section should therefore not be read as an estimate of a single channel.

2.3 The gross profit margin

Let q_t be the home-currency value against the US dollar and write the gross profit margin as $PM_t = (S_t - C_t)/S_t$. A dollar shock moves both trade-weighted rates, $d \ln q_t^X = b^X d \ln q_t$ and $d \ln q_t^M = b^M d \ln q_t$, where $b^X > 0$ and $b^M > 0$ is assumed. Substituting into Equation (5) and writing the cost share as $C_t/S_t = 1 - PM_t$ gives the model-implied operating component of the dollar currency beta:

$$\beta_q^{CF,LR} = \Lambda \frac{S_t}{P_t} \left[-\theta^X b^X + (1 - PM_t) \theta^M b^M \right], \quad (8)$$

Holding the other terms fixed, its margin derivative is

$$\frac{\partial \beta_q^{CF,LR}}{\partial PM_t} = -\Lambda \frac{S_t}{P_t} \theta^M b^M < 0 \quad \text{if } b^M > 0. \quad (9)$$

The export term lowers the operating beta and does not depend on the margin. The import term raises that beta, because profit rises after an appreciation against import origins. The margin enters as a multiplier on this import term. A high-margin economy has a relatively smaller cost channel and a more negative operating beta, while a low-margin economy weights the import term more heavily, so the currency beta can be positive. This difference can generate the difference in signs shown in Figure 1. The empirical section formally tests this prediction.

2.4 Dollar invoicing

A large share of world trade is invoiced in US dollars (Gopinath, Boz, Casas, Díez, Gourinchas, and Plagborg-Møller, 2020; Boz, Casas, Georgiadis, Gopinath, Le Mezo, Mehl, and Nguyen, 2022). When an invoice is fixed in dollars, the home-currency value of the transaction moves with the firm's dollar rate rather than with the bilateral rate against its trading partner. On the cost side, dollar invoicing breaks the link between the import-weights in the short-run, because the home-currency unit cost of a dollar-invoiced input depends on q and not on q^M .

On the revenue side, dollar invoicing may not reduce the effect of the bilateral exchange rate shock. Holding the dollar rate fixed, the foreign buyer's price moves with the bilateral rate regardless of the invoice, so export quantity and export revenue respond to q^X with the same elasticity for any dollar share, evaluated at the exchange rate when the invoice was written. A weaker revenue response at higher dollar invoicing would require an additional friction, such as export quantities that are slow to adjust to the buyer's price.

Invoice prices reset within about a year (Gopinath and Rigobon, 2008), while equity value weights the entire future path. Suppose invoice prices are fixed for H periods and reset thereafter, so that expected profit is $(1+g)^s \Pi_t^{SR}$ for $s < H$ and $(1+g)^s \Pi_t^{LR}$ for $s \geq H$, where Π_t^{SR} and Π_t^{LR} are profit at the sticky and at the flexible prices. Summing the two geometric series in Equation (4),

$$P_t = \omega_{SR} \Pi_t^{SR} + \mathcal{M} \Pi_t^{LR}, \quad \omega_{SR} = \frac{1 - \gamma^H}{1 - \gamma}, \quad \mathcal{M} = \frac{\gamma^H}{1 - \gamma}, \quad (10)$$

At the contracting-date baseline, where $\Pi_t^{SR} = \Pi_t^{LR}$, the sticky window has a share $1 - \gamma^H$ of the valuation weight. For example, with $\delta = 0.95$, $g = 0.03$, and $H = 1$ (year), this weight is about 2%. If long-run and short-run discounted profits have comparable effects on total equity prices, the effect of invoicing should be more evident in annual revenue and costs than in equity returns.

2.5 Empirical roadmap

The framework yields four empirical implications that guide the analysis below. First, equity returns should react negatively to export-destination appreciations and positively to import-origin appreciations. Second, these sensitivities should scale with industry trade intensity. Third, dollar invoicing could weaken cost and revenue responses in annual accounting data while the long-run response and equity exposure should remain unchanged. Finally, cross-country and cross-industry variation in gross profit margins should help explain the sign and magnitude of aggregate USD currency betas.

3 Data

3.1 Data sources

OECD Inter-Country Input-Output Tables (ICIO) provide bilateral trade flow data between 1995 and 2022, which report inter-country and inter-industry flows in basic prices and current values for both intermediate and final goods and services. Gross fixed capital formation (GFCF) and changes in inventories (INVNT) are excluded from the trade flow measures, because capital formation reflects one-time investment spending rather than recurring trade

relationships, and inventory changes are volatile and often negative, introducing noise into the trade weights. In addition, for trade inflows, final consumption expenditures by households (HFCE), non-profit institutions serving households (NPISH), and general government (GGFC) are excluded in the calculation. This is because only intermediate goods and services, used in production, directly influence a corporation’s costs and profitability. However, since they are part of firm’s revenue, for export-volume weighted currency returns, these trade flows are included. All goods and services exported contribute to the revenues of firms in a focal country, regardless of whether they are for final consumption or intermediate use by end-users abroad. The ability to disaggregate trade flows by intermediate use and final demand category is a unique feature of the ICIO database.

As an alternative source of bilateral trade flows at the country level, the International Monetary Fund (IMF) Direction of Trade Statistics (DOTS) database is also used. DOTS provides comprehensive country-to-country trade data with broader country coverage and a longer historical time series than the OECD ICIO. However, DOTS differs from the ICIO in two important dimensions. First, it is based on customs records rather than national accounts, and it covers merchandise trade only, excluding services. Second, it does not offer industry-level disaggregation, so it is used exclusively for country-level analysis.³

Financial data is obtained mainly from a combination of Compustat North America and Compustat Global. The dependent variables are stock returns at the country-industry and country levels. For country-industry returns, firms are assigned to countries by headquarter location and mapped from Compustat SIC codes to ISIC Rev. 4, with individual returns aggregated into equally- and value-weighted monthly portfolios. Requiring markets to have

³Import values in DOTS are recorded CIF (cost, insurance, and freight). CIF values are converted by applying a 6% CIF-FOB margin as derived from the OECD International Transport and Insurance Costs database (Marini, Dippelsman, and Stanger, 2018).

at least 100 firms for at least 60 monthly observations yields 30 countries and 36,248 unique firms over the 1996–2023 sample period, reported in Table 1. Country-level regressions use MSCI total-return indexes benchmarked against the MSCI World Developed index. In specifications using returns in excess of the US, the United States (the benchmark) and Hong Kong (pegged to USD) are excluded.

Monthly exchange rates are from Compustat Global. For pairs not covered by Compustat, it is augmented with the values of Datastream. For each currency c , the source series records its USD value, in US dollars per unit of local currency. Monthly USD currency returns are constructed from changes in this rate and denoted by $\Delta q_{c,t}$. Thus, a positive value of $\Delta q_{c,t}$ denotes an appreciation of country c 's currency against the US dollar, and a negative value denotes a depreciation.

Firm-level annual financial statement data — revenue and cost of goods sold (COGS) — are also from Compustat database. Accounting values are reported in local currency, and each observation is assigned to its firm's reporting year. Firm-year observations with negative or missing revenue and COGS are filtered. Each firm is mapped to ICIO industries via its reported NAICS code, mapped to ISIC Rev. 4. When a NAICS code corresponds to more than one ISIC industry, the firm's revenue and COGS are divided equally across them. The country-industry profit margin is the ratio of country-industry sums. The profit margin at the country-industry level is defined as:

$$PM_{c,i,t} = \frac{\text{Revenue}_{c,i,t} - \text{COGS}_{c,i,t}}{\text{Revenue}_{c,i,t}}, \quad (11)$$

where the Revenue and COGS values are the sum of the values of the individual firms. The aggregate country profit margin is defined analogously.

For annual accounting regressions, monthly country and trade-partner currency returns are compounded over the year to construct the export- and import-weighted annual currency returns as the difference between the focal country’s return and its trade-weighted partner return both against USD. Only country-industry-year cells with aggregate USD market capitalization of at least \$50 million are kept in the analysis to avoid micro-cap firms substantially affecting the panel regression estimates, which is equally weighted. The log growth rate of the revenue and COGS are capped to be within -50% and 50% range, and changes in profit margin are winsorized at the 5th and 95th percentiles. The share of exports and imports invoiced in US dollars ($INV_{c,t}^X$ and $INV_{c,t}^M$) is taken from Boz, Casas, Georgiadis, Gopinath, Le Mezo, Mehl, and Nguyen (2022), which is forward and backward filled.

Country-level financial controls combine several external sources. Net international investment positions (NIIP) are from the IMF’s International Investment Position statistics. Foreign-currency corporate debt ($CorpDebt$) is from the BIS international debt statistics, while gross foreign-currency liabilities ($GrossLib$) and net foreign-currency exposure ($NetFX$) are from the enhanced cross-border currency-exposure dataset of Bénétrix, Gautam, Juvenal, and Schmitz (2019). Foreign direct investment, stock-market capitalization, and balance-of-payments measures are obtained from the World Development Indicators of Worldbank.

3.2 Trade-weighted currency returns

Currency returns of individual countries and country-industries are measured in several ways. First, the currency return of country c relative to the USD is denoted by $\Delta q_{c,t}$, where a higher value indicates a relative appreciation of country c ’s currency against the US dollar. In

addition to the currency returns against USD, import- and export-volume weighted currency returns are also constructed for each country and country-industry. Denoting the volume of flow from country-industry c, i to country d as $v_{c,i,d}$ and the input volume of flow originating from country d as $u_{c,i,d}$, these bilateral currency returns are:

$$\Delta q_{c,i,t}^X = \Delta q_{c,t} - \sum_{d \in \Omega_{c,i}^X} \nu_{c,i,d} \Delta q_{d,t} \quad (12)$$

and

$$\Delta q_{c,i,t}^M = \Delta q_{c,t} - \sum_{d \in \Omega_{c,i}^M} \omega_{c,i,d} \Delta q_{d,t}, \quad (13)$$

where $\Omega_{c,i}^X$ and $\Omega_{c,i}^M$ are the set of export and import partners of country-industry (c, i) , respectively, and the weights $\nu_{c,i,d} = \frac{v_{c,i,d}}{\sum_{j \in \Omega_{c,i}^X} v_{c,i,j}}$, and $\omega_{c,i,d} = \frac{u_{c,i,d}}{\sum_{j \in \Omega_{c,i}^M} u_{c,i,j}}$ are the shares of trade with each partner d . In computing these currency returns, flows originating from and flowing into an industry of the same country as the focal country are excluded. In addition, countries sharing a common currency are treated as a single domestic area. Trade flows between euro-area members after the Euro formation, for instance, are excluded from the partner sets.⁴ Trade weights $\nu_{c,i,d}$ and $\omega_{c,i,d}$ are taken from the ICIO table of the prior calendar year.

Higher values of $\Delta q_{c,i,t}^X$ and $\Delta q_{c,i,t}^M$ imply a strengthening currency of the focal country relative to its export destinations and import origins, respectively. The export-to-import currency return is defined as:

$$\Delta q_{c,i,t}^{XM} = \Delta q_{c,i,t}^M - \Delta q_{c,i,t}^X. \quad (14)$$

⁴The Appendix reports results under an alternative construction that treats each country as a distinct trading partner regardless of currency union membership. The findings are qualitatively unchanged.

A higher value implies that export destination currencies appreciate relative to currency values of import origins.

3.3 Summary statistics

Table 1 summarizes the statistics used for the analysis. Panel A reports country-level means of monthly value-weighted stock returns (R^{VW}), currency returns (Δq , Δq^X , and Δq^M), and trade openness across the 30 sample countries. Panel B summarizes the means and standard deviations of the variables used in the analysis, broken down by emerging and developed economies at both the country and country-industry levels. Mean equity and currency returns are comparable across the two groups. USD currency betas are positive on average in emerging markets and negative in developed markets, although, as Figure 1 shows, the sign varies within both groups.

The export-weighted and import-weighted currency returns $\Delta q_{c,i,t}^X$ and $\Delta q_{c,i,t}^M$ are strongly correlated with each other, which has a pairwise correlation of 0.89 (not reported in the table), as both capture a relative appreciation of the domestic currency against trading partners. This collinearity reflects the geographic overlap between a country's export destinations and import origins. The partial correlation between the two measures after controlling for USD currency returns remains at 0.73. The export-to-import return is the difference between these the two measures, which removes the common domestic component. The regressions below use this differential as an alternative.

4 Empirical Results

The analysis begins at the country-industry level, where industry-specific trade patterns provide the granularity needed to separate export and import dynamics. Country-level results follow.

4.1 Stock return and currency fluctuations

The main prediction of the framework is that stock prices react negatively if the home currency increases relative to its export partners and positively if it appreciates relative to its import partners. This is first tested using country-industry panel regressions, with local-currency portfolio returns as the dependent variable. The baseline specification is

$$R_{c,i,t} = \alpha_i + \beta_X \Delta q_{c,i,t}^X + \beta_M \Delta q_{c,i,t}^M + \delta R_{c,t} + \eta R_{WLD,t} + \epsilon_{c,i,t}, \quad (15)$$

where $\Delta q_{c,i,t}^X$ and $\Delta q_{c,i,t}^M$ are the export- and import-weighted currency returns, $R_{c,t}$ is the country index return, and $R_{WLD,t}$ is the World MSCI return. In all specifications, α_i is country-industry specific. In some specifications, country-year fixed effects are added to control for slow moving country-specific variation in the macroeconomic condition.

The two bilateral currency measures are highly correlated because export destinations and import origins often overlap. Therefore, an alternative specification uses the difference between the two measures, removing the common component and is not subject to the multicollinearity between the separate export- and import-weighted returns:

$$R_{c,i,t} = \alpha_i + \beta_{XM} \Delta q_{c,i,t}^{XM} + \gamma_{USD} \Delta q_{c,t} + \delta R_{c,t} + \eta R_{WLD,t} + \epsilon_{c,i,t}, \quad (16)$$

where $\Delta q_{c,i,t}^{XM} = \Delta q_{c,i,t}^M - \Delta q_{c,i,t}^X$ measures the appreciation of export-destination currencies relative to import-origin currencies.

Table 2 describes the results of this panel regression. Panel A uses the export- and import-weighted bilateral currency returns as independent variables. Panel B presents the alternative specification using the export-to-import currency return and USD currency returns. Each panel reports results for both value-weighted (left columns) and equally-weighted (right columns) country-industry returns.

Across both panels, the signs are consistent with the theoretical prediction. The value-weighted return result in Panel A shows that a one-percentage-point appreciation relative to export destinations is associated with a 6–10 basis-point decline in monthly country-industry returns, while an equally sized appreciation relative to import origins is associated with a 9–13 basis-point increase. Comparing the magnitude with equal weighting, the import coefficients are generally larger under equal weighting relative to the export channel.

This difference implies that smaller firms are more exposed to the import cost channel and less to export channel compared to the largest firms in the economy. It is possible that larger firms export a higher fraction of their output while smaller firms import more of their input. It may also be easier for larger firms to hedge their operation. For example, larger firms may have the capability to use financial derivatives to hedge currency risk, or as Amiti, Itskhoki, and Konings (2014) show, apply a natural operational hedge to reduce their exposure. Smaller firms, conversely, often rely on imported intermediate inputs but lack access to financial hedging.

Panel B replaces the two correlated bilateral measures with $\Delta q_{c,i,t}^{XM}$ and controls for the USD return. The coefficients on $\Delta q_{c,i,t}^{XM}$ are all positive and statistically significant except for

one. This pattern is consistent with an appreciation of export-destination currencies relative to import-origin currencies being associated with higher stock returns after controlling for aggregate currency and equity-market movements. In contrast, the USD return coefficient is statistically indistinguishable from zero in most specifications. It is positive and marginally significant only in the two equally-weighted columns that control for the contemporaneous country index return, where the coefficient remains small. The results are similar when euro-area member countries are treated as distinct trading partners rather than as a single currency area (Appendix Table A1).

Generally, controlling for the contemporaneous country index return $R_{c,t}$ weakens the estimated trade channels. When controlling for country stock returns, the bilateral currency coefficients capture the additional industry-specific exposure associated with trade-partner composition, beyond the common country-wide effect. The weaker results with country controls therefore indicate that the bilateral currency shocks have both a country-level component and a residual industry-level component.

4.2 Currency beta and trade exposure

Another prediction is that the magnitude of the bilateral currency channels should depend on trade intensity. This prediction is tested using an interactive panel regression

$$\begin{aligned}
R_{c,i,t} = & \alpha_{c,i} + \beta_X \Delta q_{c,i,t}^X + \beta_M \Delta q_{c,i,t}^M + \lambda \%Trd_{c,i} \\
& + \iota_X \Delta q_{c,i,t}^X \times \%Trd_{c,i} + \iota_M \Delta q_{c,i,t}^M \times \%Trd_{c,i} + \mathbf{Controls}_{c,i,t} + \epsilon_{c,i,t},
\end{aligned} \tag{17}$$

where $\%Trd_{c,i}$ denotes the average ratio of exports plus imports divided by total output plus total input for country-industry (c, i) .

Table 3 reports these country-industry interaction regressions. Panel A presents specifications using export- and import-weighted currency returns. Across all eight columns, the export interaction ι_X is negative and statistically significant, and the import interaction ι_M is positive and highly statistically significant. The standalone linear terms β_X and β_M represent the extrapolated currency sensitivity for firms with zero trade exposure. The trade-share coefficient is absorbed by the country-industry fixed effects included in all columns. The panel shows that once country equity returns are controlled for in columns (3)–(4) and (7)–(8), the currency exposure is close to zero and statistically insignificant.

Evaluating the total marginal effect, at a low trade share of $\%Trd = 5\%$ (near the 5th percentile of the sample), the net sensitivity is economically negligible (e.g., $+0.011$ for exports and -0.005 for imports in column 4). As trade intensity rises toward the sample mean ($\%Trd \approx 21\%$), the net export sensitivity reaches -0.079 and the net import sensitivity $+0.108$, close to the magnitudes in Table 2. For highly open sectors ($\%Trd = 30\%$, near the 75th percentile), sensitivities expand further to -0.130 and $+0.172$, respectively. Panel B also suggests a strong positive and statistically significant interaction between $\Delta q_{c,i,t}^{XM}$ and $\%Trd_{c,i}$ in every specification. Taken together, the results are consistent with bilateral currency sensitivities increasing with country-industry trade intensity.

The interaction term may reflect differences among industries within countries, differences across countries, or both. The analysis studies the cross-industry dimension by comparing the import-export sensitivity differential for tradable and non-tradable industries and for goods and services. When splitting the sample by tradability, each country-industry's trade share is first averaged over time. An industry is classified as tradable when this average exceeds the median of the country-industry averages within the same country. Therefore, the classification is country-specific and time-invariant.

Figure 2 summarizes the difference in the coefficients across these two groups. In both comparisons, the differential is larger for the more trade-intensive group tradable industries and goods sectors than for non-tradable industries and service sectors. This result shows that a non-negligible fraction of the cross country-industry interactions reported in Table 3 is driven by the variation across industries.

Figure 3 examines the variation across the country dimension. For each country, it estimates the country-industry return regression separately and plots $\hat{\beta}_{M,c} - \hat{\beta}_{X,c}$ against the country's aggregate trade openness. The positive relationship in the figure shows that the export–import sensitivity differential is larger in more open economies, extending the country-industry evidence to cross-country variation. The relationship is not driven by a single outlier. The Spearman rank correlation is 0.60, close to the Pearson correlation of 0.65, and it depends only on the countries' ranks, not on the size of any country's differential. It stays between 0.56 and 0.69 when each country is dropped in turn. The appendix provides table-based counterparts to both figures.

Taken together, these results show that the country–industry interaction in Table 3 reflects both dimensions of the data. Trade-intensive industries have larger currency sensitivities within countries, and the sensitivity differential is larger across more-open countries. The interaction is therefore not driven solely by either cross-industry or cross-country variation, but is a combination of the two.

4.3 The operating channel: profit margins, revenue, and costs

This section looks for more direct evidence of the export-revenue and import-cost channels using accounting data. Firm revenue, cost of goods sold (COGS), and gross profit margins

are aggregated to the country-industry-year level, as described in the data section, and their changes are related to the bilateral currency returns defined at that level.

Table 4 reports the results of the panel regressions for annual ($h = 0$) and four-year cumulative ($h = 3$) changes in operating outcomes. In all panels, odd-numbered columns include only the contemporaneous shock, while even-numbered columns add one-year lags. All specifications control for the lagged outcome, contemporaneous and lagged global aggregate outcomes, and country $\times$ industry fixed effects.

Panel A studies the influence on gross profit margin of the trade-weighted currency returns, with one specification using export- and import-weighted shocks separately, and the second with the export-to-import differential. The panel shows that the gross profit margin is lower when the local currency appreciates relative to export destinations, whereas an appreciation relative to import origins raises it. However, the effect is not instantaneous and appears with a lag and is a longer-period effect. The export-to-import differential produces a similar pattern. Thus, the margin responses are consistent with revenue and cost channels moving in the predicted opposite directions.

Panels B and C investigate the revenue and cost channels separately, by changes in log revenue to the export-weighted returns ($\Delta q_{c,i,t}^X$) in Panel B and changes in log COGS to the import-weighted currency returns ($\Delta q_{c,i,t}^M$), respectively. Looking separately at revenue and costs, Panel B finds no one-year revenue response in the contemporaneous or lagged term, while Panel C finds no contemporaneous cost response but a significant one-year lag. At the four-year cumulative horizon, the estimated patterns are clearer. Export-relative appreciation is associated with lower cumulative revenue both contemporaneously and with

a one-year lag, while import-relative appreciation is associated with lower cumulative COGS both contemporaneously and with a lag.

Figure 4 further explores the dynamic paths using the local projection method of Jordà (2005). Panel (a) shows that the revenue response to the export-weighted shock remains negative, and Panel (b) indicates that the COGS response to the import-weighted shock is also negative, consistent with lower import costs after an import-relative appreciation. Panel (c) reports the corresponding cumulative profit-margin responses. The figure shows that both the negative export-revenue effect and the favorable import-cost effect persist over multiple horizons.

4.4 Currency invoicing and the revenue-cost channel

As discussed in the model section, high USD invoicing share can attenuate the pass-through of bilateral exchange rate shocks to firm revenue and costs (Gopinath, Boz, Casas, Díez, Gourinchas, and Plagborg-Møller, 2020). This prediction is tested by interacting the export-weighted shock $\Delta q_{c,i,t}^X$ with the share of exports invoiced in US dollars, $INV_{c,t}^X$, in the revenue regressions. The COGS regressions analogously interact the import-weighted shock $\Delta q_{c,i,t}^M$ with the USD share of imports, $INV_{c,t}^M$. The sample sizes differ across panels due to differences in international reporting coverage for export and import invoicing shares. A positive interaction coefficient indicates attenuation of the operating channel at a higher USD invoicing share.

Table 5 reports the results at one-year ($h = 0$) and four-year cumulative ($h = 3$) horizons. In both panels, columns (1) and (3) include the contemporaneous interaction, while columns (2) and (4) also include its one-year lag. Both panels confirm the baseline effect, that

an export-relative appreciation lowers revenue, and an import-relative appreciation lowers COGS.

For the interaction terms, the evidence is concentrated at the one-year horizon. For revenue, the contemporaneous and the one-year lagged interaction is individually positive and statistically significant, showing that invoicing protection delays pass-through into subsequent periods. For COGS, the contemporaneous interaction is positive and statistically significant in both one-year specifications. For both revenue and COGS, the sum of the contemporaneous and lagged interactions is positive and highly statistically significant, summarized in the last row just above the fixed effects.

At the four-year cumulative horizon, however, neither the individual nor the summed interaction coefficients are different from zero. This evidence is consequently consistent with invoicing affecting accounting variables temporarily rather than permanently. An appendix figure further tracks how these attenuation effects evolve over a multi-year horizon. This short-horizon pattern is consistent with Gopinath and Rigobon (2008), who document median invoice price duration is around a year, with 10.6 months for U.S. imports and 12.8 months for exports.

5 Country-level analysis

5.1 Country-level stock returns and bilateral currency shocks

This section studies whether the logic applied to country-industry also applies at the country level. Although aggregating across industries may obscure some of the granular dy-

namics, a country's overall trade relationships should still affect stock index returns through changes in currency value relative to its aggregate trading partners.

In this section, MSCI country-index returns are regressed on country-level export- and import-weighted currency returns, constructed after aggregating the ICIO sample to the country level. The corresponding DOTS measures directly use aggregate bilateral trade flows. Dependent variables include both raw returns and returns in excess of the US index return. The raw returns specification controls for world equity returns $R_{WLD,t}$, while excess-return specification omits them, following the benchmark-relative specification in Hau and Rey (2006):

$$R_{c,t} = \alpha_c + \beta_X \Delta q_{c,t}^X + \beta_M \Delta q_{c,t}^M + \eta R_{WLD,t} + \epsilon_{c,t}, \quad (18)$$

$$R_{c,t} = \alpha_c + \beta_{XM} \Delta q_{c,t}^{XM} + \gamma_{USD} \Delta q_{c,t} + \eta R_{WLD,t} + \epsilon_{c,t}.$$

The trade-weighted currency returns computed after aggregation raise the correlation between the export- and import-weighted returns to 0.989 in both the ICIO and DOTS samples. The export-to-import return ($\Delta q_{c,t}^{XM}$), therefore, provides a better identified measure of the net trade channel.

Table 6 describes the corresponding country-level pattern. Panel A summarizes the result for the ICIO sample, and Panel B is for the DOTS sample. Even-numbered columns include country $\times$ year fixed effects. In both samples, the export-weighted coefficient is negative and the import-weighted coefficient positive, and each is statistically significant for both raw and excess country returns despite the two series being highly correlated. The differential specification, which is not subject to multicollinearity is likewise positive and statistically significant in every column.

If the balance sheet effect of foreign currency debt is an important channel, γ_{USD} is likely to be meaningful. International debt is denominated mainly in US dollars rather than in the currency baskets of a country’s export destinations and import origins, so a balance-sheet effect would load primarily on the bilateral USD return rather than on the export-to-import differential. If indeed the US is the net debtor and the rest of world is a net creditor, we expect a negative sign on γ_{USD} . The panel shows that the bilateral USD-return coefficient in the differential specifications is statistically insignificant. These results show that the country-level evidence is consistent with the export-revenue and import-cost channels identified in the country-industry panel, rather than with a single common USD exposure.

Notably, the country-level coefficients are roughly an order of magnitude larger than their country-industry counterparts in Table 2. This difference principally reflects scaling rather than different economic content. First, the country-industry specifications control for country index returns and thereby remove the common component of the currency shock. The preceding table showed that the country index control reduces the estimated differential coefficient by roughly one-third. Second, aggregation compresses the dispersion of the shock. Export destinations and import origins overlap more closely at the country level than within individual industries. Therefore, the standard deviation of the export-to-import returns is 0.32 percentage points at the country level, compared with 1.07 percentage points at the country-industry level. Hence, evaluated at a one-standard-deviation shock, the estimated return effects are therefore approximately 10 basis points at the country-industry level and 26 basis points at the country level.⁵

⁵For each level of aggregation, the calculation is $\hat{\beta}_{XM} \times \text{sd}(\Delta q^{XM})$. Comparing Table 2, Panel B, column (1) and Table 6, Panel A, column (3), yields 10 and 26 bp, respectively. Part of the remaining gap may also reflect the different return series, as country-industry returns are built from Compustat portfolios whereas country returns are MSCI index returns.

5.2 Aggregate net export

Net export is a useful descriptive summary of a country's net position in the two bilateral channels. Table 7 tests this channel by interacting USD currency returns with lagged net exports and, separately, export and import shares. Net export (NX), import share (%IMP), and export share (%Exp) are defined as:

$$\begin{aligned}NX_{c,t}^- &= (EXP_{c,t}^- - IMP_{c,t}^-)/(EXP_{c,t}^- + IMP_{c,t}^-) \\ \%IMP_{c,t}^- &= IMP_{c,t}^-/D_{c,t}^- \\ \%EXP_{c,t}^- &= EXP_{c,t}^-/D_{c,t}^-, \end{aligned}$$

where $D_{c,t}$ is the sum of input and output volume for ICIO and Total GDP for DOTS. Throughout, a superscript $-$ denotes the value of a country characteristic in the year preceding year t , so that it is measured before the currency and stock returns it is interacted with.

In the ICIO sample, the net-export interaction is negative and statistically significant for both raw and U.S.-excess returns. The import-share interaction is positive and likewise statistically significant in both return definitions, whereas the export-share interaction has a negative sign as predicted but is statistically insignificant. In the DOTS sample, all three interaction coefficients have signs consistent with theory, but none is statistically significant. Thus, the aggregate trade measures are directionally consistent with the bilateral mechanism but provide only weak supplementary evidence.

The weak statistical evidence found consistently in these aggregate specifications likely helps explain why prior studies often find little or no evidence of trade being a determinant

of currency risk exposure. Aggregate net-trade and trade-share measures collapse heterogeneous industry and trading-partner exposures into a single exposure, while a common USD return combines export-revenue and imported-input-cost effects that work in opposite directions. Both aggregation and offsetting make the bilateral trade channel difficult to identify at the country level.

5.3 Gross profit margin and currency beta

The model predicts that the revenue channel should be relatively more important in high-margin economies, producing a more negative USD currency beta, and the cost channel should dominate for low-margin countries. This prediction is tested by interacting the USD currency return with lagged country profit margins, while controlling for price-to-sales, net exports, NIIP, and other external-finance variables. The regression considered is:

$$R_{c,t} = \alpha_c + (\beta + \gamma_{PM} PM_{c,t}^- + \gamma' \mathbf{Z}_{c,t}^-) \Delta q_{c,t} + \eta R_{WLD,t} + \epsilon_{c,t}, \quad (19)$$

where $\mathbf{Z}_{c,t}^-$ contains the additional lagged country characteristics. All specifications include their linear terms and country fixed effects.

Table 8 reports this regression using raw country index returns in columns (1)–(4) and stock returns in excess of U.S. stock returns in columns (5)–(8). Panel A uses the ICIO sample, and Panel B uses the DOTS sample. Across all specifications, the margin interaction term is negative. This negative interaction could reflect country characteristics correlated with both profit margins and currency exposure. To address this concern, the next columns consider a set of progressive controls, including price-to-sales, which captures the valuation of profits relative to sales, and additional country characteristics (net exports, NIIP) and

detailed financial exposures (corporate FX debt, gross liabilities, and net foreign assets). Specifically, the last set of controls account for the financial balance-sheet channel, through which foreign-currency liabilities can affect equity returns independently of trade exposure (Kim, Tesar, and Zhang, 2015; Bruno and Shin, 2017; Kalemli-Ozcan, Liu, and Shim, 2021). The two panels show that for all controls considered, the profit-margin interaction is negative and statistically significant in both samples. The margin effect is therefore not a representation of different financial exposures.

The baseline estimate of γ_{PM} depends on both differences in profit margins across countries and changes within countries over time. To distinguish these sources of variation, I decompose each unit's profit margin into its time-series mean, $\overline{PM}_c^-$ and its deviation from that mean $\widetilde{PM}_{c,t}^-$, so that

$$PM_{c,t}^- = \overline{PM}_c^- + \widetilde{PM}_{c,t}^-,$$

and each of the two components are interacted separately with currency returns.

Table 9 shows the results of the regression with the interactive terms entered separately. Panel A shows the country-level regressions and Panel B the country-industry-level regressions. The specifications in columns (3) and (6) include a separate USD currency slope for each country in Panel A and for each country-industry unit in Panel B. These slopes absorb the between-unit margin interaction, so the reported interaction coefficient is for the within-unit margin variation.

In Panel A, both components negatively affect the currency beta. Between the two, the between-country interaction is roughly three times larger than the within-country interaction. Both are statistically significant in the decomposition. When country-specific FX slopes are included in columns (3) and (6), the interactive coefficients remain negative but become

statistically insignificant. Panel B reproduces the pattern at the country-industry level. The within-country-industry interaction is negative for both equally and value-weighted returns. Adding country-industry fixed effects and allowing every country-industry unit its own currency slope leaves the estimate essentially unchanged, so the association does not entirely arise from differences in average exposure across industries.

To examine whether the profit-margin association is concentrated in trade-intensive economies, Table 10 reports regressions testing the triple interaction between the currency return, the profit margin, and aggregate trade intensity ($\Delta q_{c,t} \times PM_{c,t}^- \times \%TRD_{c,t}^-$) without additional control variables. In both the ICIO (Panel A) and DOTS (Panel B) samples, the triple interaction term is negative and highly statistically significant across both raw returns (Column 4) and excess returns (Column 8).

Figure 5 provides the cross-sectional scatterplot of country USD currency betas against aggregate gross profit margins. Panel (a) shows that the association is economically substantial, with a cross-country correlation of -0.55 and a rank correlation of -0.59 across 28 countries. This pattern does not change after controlling for additional country-level variables, namely the net export, net investment position, corporate foreign currency debt variables considered earlier. Thus, the relationship to gross profit margins accounts for approximately 30% of the cross-country variation in USD currency betas. The negative slope is consistent with the prediction that higher-margin economies exhibit more negative currency betas.

5.4 Implied and realized currency betas

I next ask whether the estimated revenue and cost responses help explain cross-country differences in USD equity currency betas. Using the model's present-value decomposition, I combine these responses with each country's cost share and sales-to-price ratio to construct an implied operating beta. This measure does not use the covariance of equity and currency returns. The decomposition implies:

$$\hat{\beta}^{\text{impl}} = (S/P)V^S - (C/S)(S/P)V^C, \quad (20)$$

where the revenue and cost elasticities (V^S and V^C) are estimated separately in country-industry regressions on the USD currency return, and the coefficients are aggregated to the country level using sales weight or value weights, and combined with the country level sales-to-price ratio and cost shares. The two elasticity terms are the discounted sum of the revenue or cost responses over a three-year horizon, with annual discount factor 0.97.

This beta is compared with the USD beta estimated from the OLS relation between annual country equity returns and the USD currency return, using only financial-market data and no accounting data. It is worth noting that the implied beta does not capture the true beta because of the fixed horizon used for the sensitivity estimation. Therefore, the scale is not directly comparable to that of the realized beta. Nevertheless, the comparison between the implied and the estimated beta is still informative about its cross-country pattern.

Figure 6 reports the comparison under market-capitalization and sales weighting. Across both panels, the figure shows that the implied and realized betas are strongly positively related, with correlations of 0.54 and 0.44. The comparison indicates that about a quarter

of the cross-country variation in USD beta is explained by this operating channel. This is not an estimate of the entire channel that affects the currency beta. Foreign-currency debt, financial hedging, and the portfolio-rebalancing and discount-rate channels emphasized by Hau and Rey (2006), Camanho, Hau, and Rey (2022), and Ben Zeev and Nathan (2024), as well as measurement error in the accounting elasticities, may account for the remaining variation.

6 Conclusion

This paper revisits the relationship between exchange rates and equity values through the bilateral trade network both at the country-industry and country level. Previous literature used a single USD currency return and/or used the trade measure aggregated to the country level. Aggregation and using a single benchmark combine two economically distinct shocks into one: an appreciation of the local currency relative to export destinations weakens export revenue, and an appreciation relative to import origins lowers imported-input costs. Because these shocks have opposing effects on stock returns, a single benchmark can obscure economically meaningful relationships.

At the country-industry level, appreciation relative to export destinations is associated with lower stock returns, whereas import-relative appreciation is associated with higher returns. Both of these associations are stronger in more trade-intensive industries. Revenue and cost data confirm the underlying mechanism. Export-relative appreciations reduce revenue, while import-relative appreciations lower costs. Their offsetting effects on profitability help explain why a single dollar benchmark is either insignificant or inconsistent.

Country-index returns and import- and export-weighted currency returns constructed from both ICIO and DOTS trade data exhibit the same signed pattern. Across countries, gross profit margins are negatively associated with USD currency betas, consistent with a relatively larger export-revenue component in higher-margin economies. Aggregate net-trade and external-balance variables are useful controls but provide less uniform independent results in the setting of this paper. A currency beta built from revenue and cost fundamentals alone has a correlation of roughly 0.5 with realized currency betas, corresponding to about one-quarter of their cross-country variation.

These results suggest that empirical studies of equity currency risk should consider exchange-rate movements against export destinations and import origins alongside a common currency benchmark. For investors and risk managers, a market's trade network and corporate profitability jointly provide more informative signals of currency sensitivity than standard USD exposure measures. For policymakers, the results imply that the equity-market consequences of exchange-rate movements depend on the composition of trade rather than on the aggregate trade weights or net exports.

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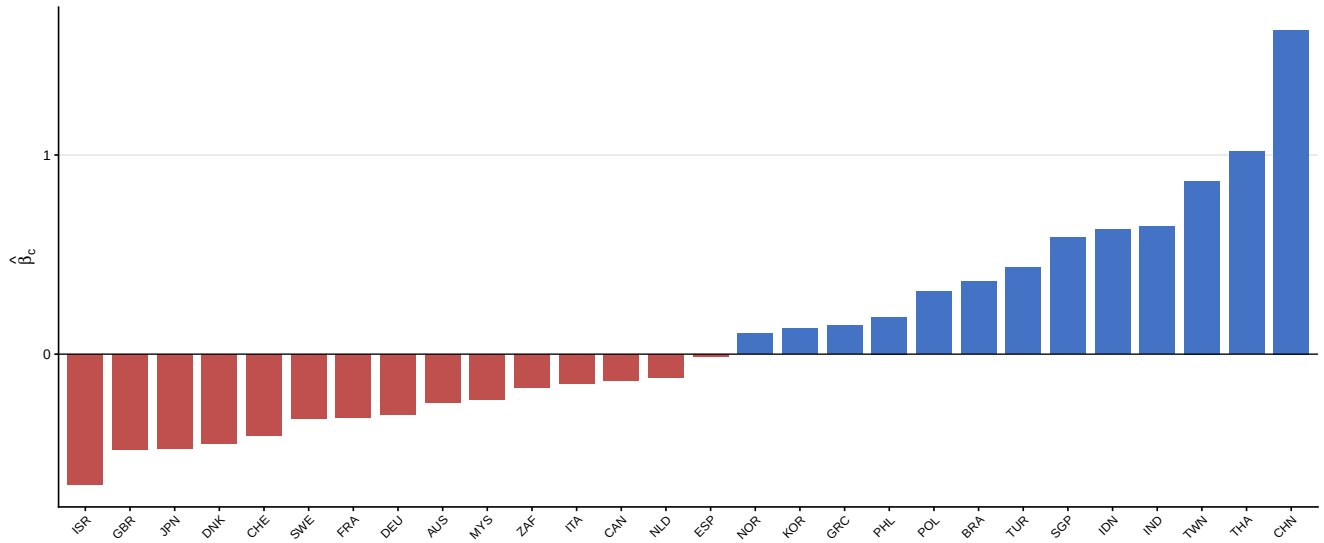


Figure 1: Currency beta of country index excess returns by country

This figure displays the slope coefficients $\hat{\beta}_c$ of the OLS regression $R_{c,t} - R_{US,t} = \alpha_c + \beta_c \Delta q_{c,t} + \varepsilon_{c,t}$, estimated country by country, where $R_{c,t}$ is country c 's MSCI equity-index return, $R_{US,t}$ is the US equity-index return, and $\Delta q_{c,t}$ is the return on country c 's currency against the US dollar. Countries are ordered by their estimated currency beta. Positive (negative) bars indicate that a local-currency appreciation is associated with higher (lower) country equity returns relative to the United States.

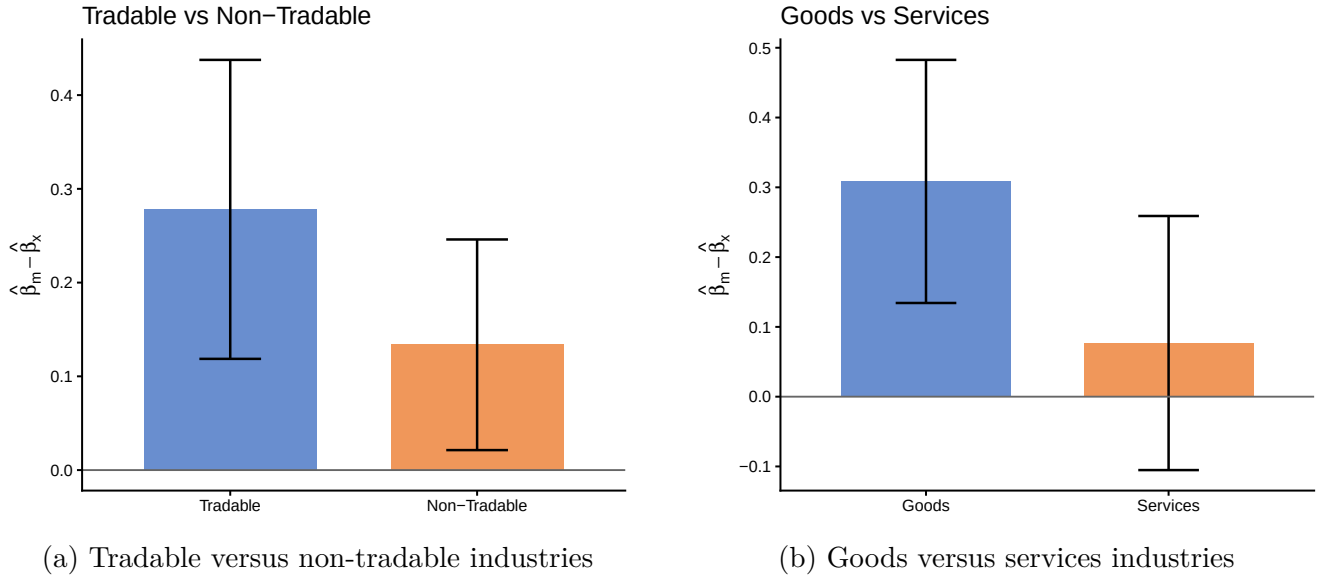


Figure 2: Currency exposure difference across industry groups

This figure shows the difference between import- and export-weighted currency coefficients, $\hat{\beta}_M - \hat{\beta}_X$, from separate country-industry stock-return panel regressions estimated for each group. Panel (a) splits industries by their time-average trade share relative to the median within each country, while Panel (b) compares goods (ISIC sections A–C) with services (the remainder). All regressions include country-industry fixed effects and the world MSCI return. Error bars are 90% confidence intervals based on Driscoll–Kraay standard errors with three lags.

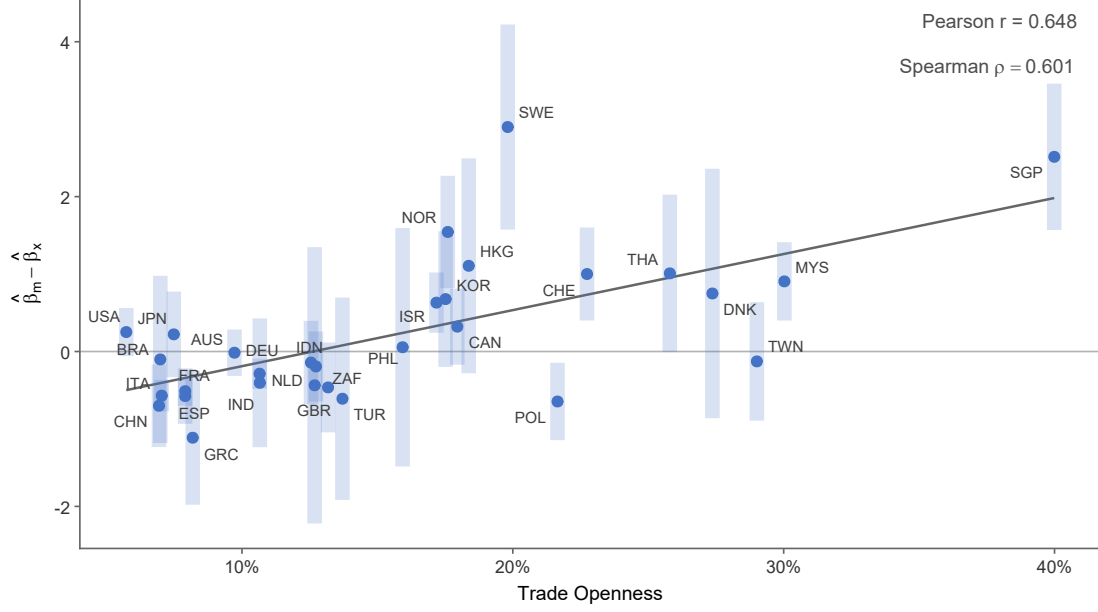
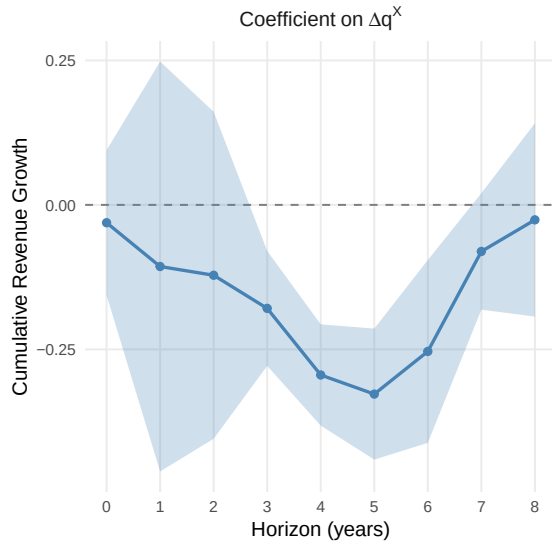


Figure 3: Currency exposure difference versus trade openness across countries

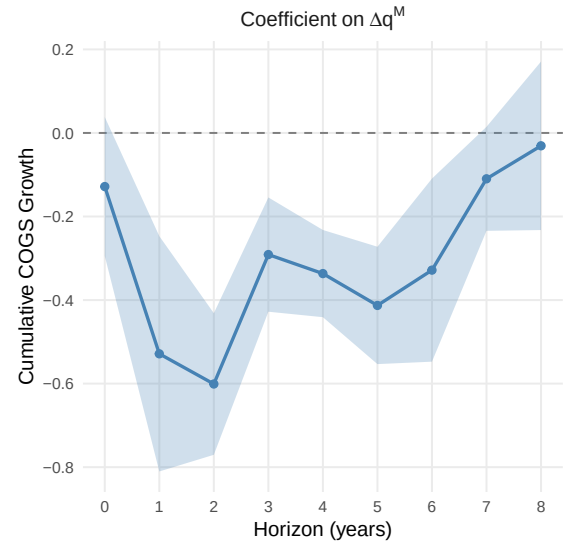
This figure plots the difference between import- and export-weighted currency coefficients, $\hat{\beta}_M - \hat{\beta}_X$, from separate industry stock-return panel regressions estimated for each country, against its long-run trade openness. The regression estimated is:

$$R_{c,i,t} = \alpha_c + \beta_{X,c} \Delta q_{c,i,t}^X + \beta_{M,c} \Delta q_{c,i,t}^M + \gamma_c R_{WLD,t} + \varepsilon_{c,i,t},$$

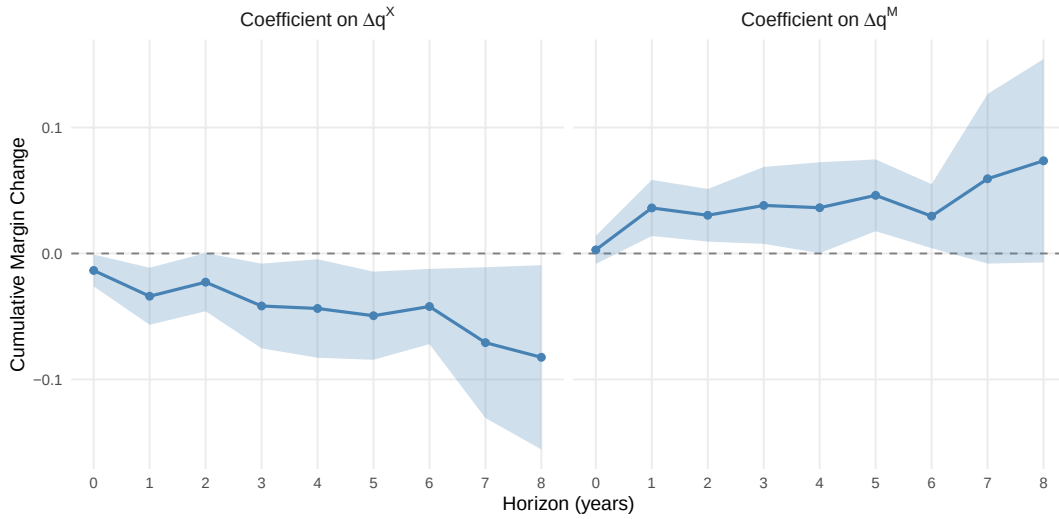
where $R_{c,i,t}$ is the value-weighted country-industry stock return, $\Delta q_{c,i,t}^X$ ($\Delta q_{c,i,t}^M$) is the focal country-industry's currency return relative to its export destinations (import origins), and $R_{WLD,t}$ is the world MSCI return. The vertical axis reports $\hat{\beta}_{M,c} - \hat{\beta}_{X,c}$, and the horizontal axis reports each country's trade openness, defined as the ratio of the country's exports plus imports to its total output plus inputs, computed from the OECD Input-Output tables and aggregated across years. The 90% confidence intervals, based on Driscoll-Kraay standard errors with three lags, are shown around each country's point estimate. The solid line is the cross-country OLS fit of the country-level estimates, and r and ρ denote the corresponding Pearson and Spearman rank correlation coefficients, respectively.



(a) Revenue growth



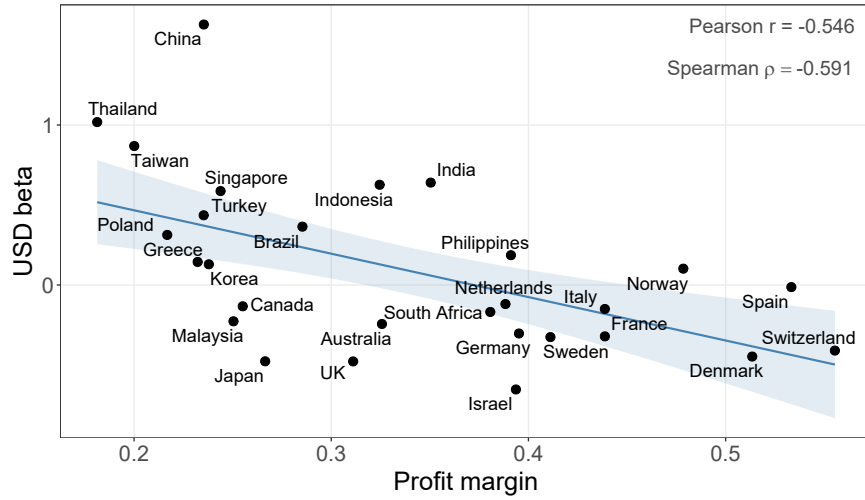
(b) COGS growth



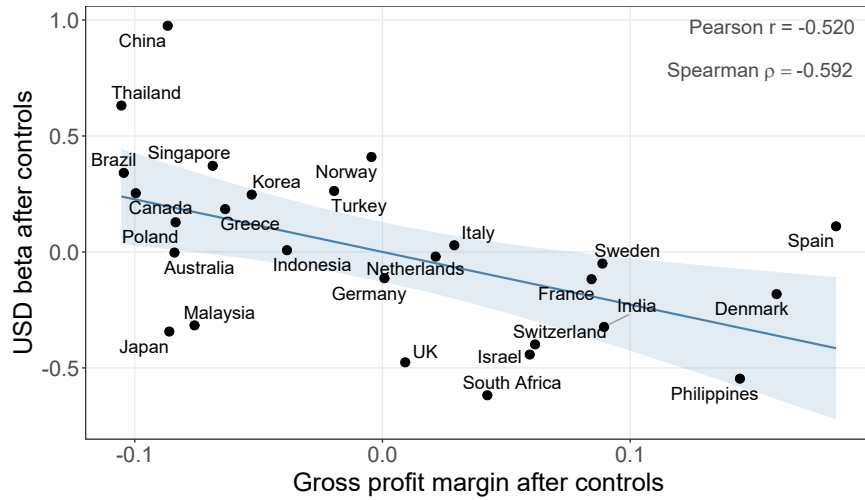
(c) Profit margin change

Figure 4: Country-industry revenue and cost responses to currency returns

This figure illustrates the cumulative responses from $t - 1$ through $t + h$ in response to currency returns between $t - 1$ to t , for $h = 0, \dots, 8$, estimated via local projections. Panels (a) and (b) plot the export-weighted currency coefficient for revenue growth and the import-weighted coefficient for COGS growth, respectively. Panel (c) plots both contemporaneous import- and export-weighted coefficients for the change in gross profit margin, defined as $(\text{Rev} - \text{COGS})/\text{Rev}$. All regressions include country-industry fixed effects, lagged currency shocks, the lagged outcome, and current and lagged global outcome controls. Shaded bands are 90% confidence intervals based on Driscoll–Kraay standard errors with three lags.



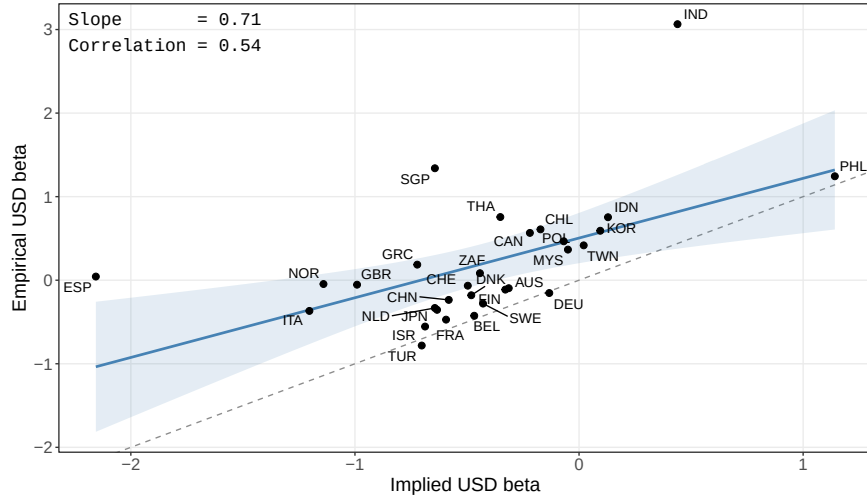
(a) Relationship without controls



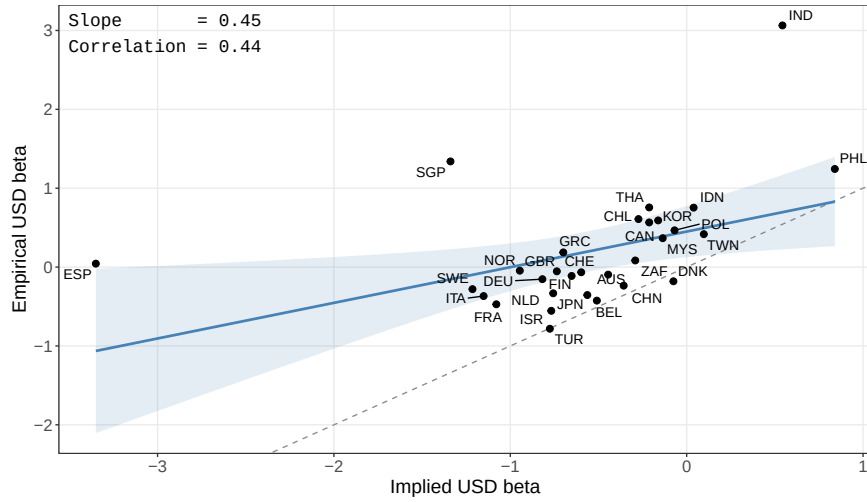
(b) Relationship after controls

Figure 5: Currency beta and gross profit margins

Panel (a) plots each country's USD currency beta, estimated as in Figure 1, against its gross profit margin aggregated from firm-level accounting values. Panel (b) plots both variables after removing the effects of country-average log PPP GDP per capita, log price-to-sales, net exports divided by total trade, NIIP/GDP, corporate foreign-currency debt/GDP, foreign-currency debt liabilities/GDP, and net foreign-currency exposure. Taiwan is excluded from panel (b) because the controls are unavailable. Solid lines are unweighted OLS fits across countries and shaded regions are 90% confidence intervals.



(a) Market-cap-weighted



(b) Sales-weighted

Figure 6: Realized and implied country USD currency betas

This figure compares each country's estimated USD currency beta with the beta implied by the decomposition of country-level revenue and cost responses. Panels (a) and (b) use market-capitalization and sales weights, respectively, to aggregate the revenue and cost sensitivities of each industry. The solid line is the cross-country OLS fit, and the reported slope and correlation summarize the correspondence between realized and implied betas. The implied beta is built from revenue and cost elasticities, sales, and cost shares, and uses no equity-return covariation.

Table 1: Summary statistics

This table reports summary statistics. Panel A reports country-level coverage, average monthly value-weighted equity returns (R^{VW}), currency returns against USD (Δq), export- and import-weighted currency returns (Δq^X and Δq^M), and trade openness, defined as the sum of exports and imports divided by the sum of inputs and outputs. N is the number of country-industry-month observations. Panel B reports means and standard deviations (in parentheses) separated by the development of the economy. Revenue, COGS, and market capitalization are in millions, millions, and billions of US dollars, respectively.

Panel A. Equity and currency returns of countries									
Country	Start	Months	Firms	R^{VW}	Δq	Δq^X	Δq^M	Openness	N
Australia	1996-01	336	515	0.94%	0.05%	0.11%	0.08%	9.7%	15,177
Brazil	2010-04	152	116	1.34%	-0.57%	-0.38%	-0.40%	7.0%	5,089
Canada	1996-01	335	568	0.89%	0.03%	0.09%	0.05%	17.9%	15,996
China	1999-01	299	1,560	0.97%	0.05%	0.12%	0.06%	6.9%	13,287
Denmark	2001-05	203	114	0.97%	0.19%	0.09%	0.06%	27.3%	5,136
France	1996-01	335	383	0.87%	-0.02%	0.22%	0.07%	7.9%	13,033
Germany	1996-01	335	411	0.65%	-0.02%	0.21%	0.07%	10.7%	14,092
Greece	2001-08	161	113	-0.06%	0.23%	0.18%	0.21%	8.2%	5,415
Hong Kong	1996-01	335	524	0.74%	-0.00%	0.07%	0.03%	18.4%	13,383
India	1997-01	324	1,129	1.94%	-0.24%	-0.14%	-0.20%	10.7%	13,950
Indonesia	2000-08	280	256	1.25%	-0.16%	-0.10%	-0.17%	12.5%	10,244
Israel	2001-03	274	207	0.84%	0.08%	0.14%	0.08%	17.2%	8,926
Italy	1996-01	335	154	0.65%	0.01%	0.27%	0.14%	7.0%	11,556
Japan	1996-01	335	2,155	0.50%	-0.06%	0.01%	0.01%	7.5%	14,886
Korea	1999-01	299	918	0.99%	0.02%	0.09%	0.06%	17.5%	12,667
Malaysia	1999-01	299	541	0.76%	-0.05%	0.01%	-0.04%	30.0%	13,532
Netherlands	1998-08	84	106	0.16%	0.51%	0.35%	0.37%	13.1%	2,565
Norway	2005-08	220	126	0.73%	-0.18%	-0.09%	-0.13%	17.6%	7,028
Philippines	2012-05	126	110	0.61%	-0.18%	-0.01%	-0.01%	15.9%	3,395
Poland	2007-08	195	252	0.50%	-0.15%	0.01%	0.03%	21.6%	7,027
Singapore	1996-01	335	325	0.57%	0.04%	0.14%	0.11%	40.0%	13,291
South Africa	1999-01	290	124	1.38%	-0.28%	-0.23%	-0.25%	12.8%	9,470
Spain	2016-04	92	130	0.20%	-0.03%	0.63%	0.14%	7.9%	2,846
Sweden	1998-06	306	268	0.78%	-0.05%	-0.01%	-0.02%	19.8%	11,013
Switzerland	1996-08	326	138	0.69%	0.16%	0.22%	0.20%	22.7%	10,748
Taiwan	1999-01	300	993	0.79%	0.03%	0.07%	0.03%	29.0%	10,978
Thailand	1999-01	299	332	1.15%	0.03%	0.10%	0.07%	25.8%	12,073
Turkey	2001-09	267	161	3.05%	-1.08%	-1.08%	-1.08%	13.7%	8,467
United Kingdom	1996-01	326	960	0.82%	-0.05%	0.03%	0.01%	13.2%	14,663
United States	1996-01	326	3,085	1.00%	0.00%	0.10%	0.06%	5.7%	15,610

Panel B. Summary statistics by emerging and developed market				
	Emerging market		Developed market	
	Country	Industry	Country	Industry
R^{VW}	0.007 (0.072)	0.012 (0.140)	0.005 (0.056)	0.007 (0.108)
Δq	-0.002 (0.030)	-0.002 (0.029)	0.000 (0.027)	0.000 (0.027)
Δq^X	-0.001 (0.026)	-0.001 (0.025)	0.001 (0.018)	0.001 (0.021)
Δq^M	-0.002 (0.026)	-0.001 (0.025)	0.001 (0.018)	0.001 (0.019)
Stock's USD beta	0.464 (0.445)	0.255 (0.564)	-0.190 (0.312)	-0.206 (0.614)
Market cap (Billion USD)	559.3 (1,083.1)	14.6 (72.2)	669.6 (726.1)	17.3 (48.0)
Profit margin	0.292 (0.140)	0.333 (0.217)	0.385 (0.158)	0.417 (0.261)
Revenue (Million USD)	344,732 (658,636)	9,700 (40,299)	538,459 (857,039)	15,028 (51,711)
COGS (Million USD)	259,816 (507,968)	7,311 (32,356)	365,214 (633,921)	10,193 (40,801)

Table 2: Baseline country-industry stock returns and exposure to trade-weighted currency returns

This table describes the baseline regression results at the country-industry level. The dependent variables are country-industry monthly equity returns. Columns (1)–(4) report value-weighted returns and columns (5)–(8) report equally weighted returns. In Panel A, the independent variables include the export-weighted currency return ($\Delta q_{c,i,t}^X$), the import-weighted currency return ($\Delta q_{c,i,t}^M$), the country index return ($R_{c,t}$), and the world index return ($R_{WLD,t}$). In Panel B, the independent variables are the export-to-import currency return ($\Delta q_{c,i,t}^{XM}$), the bilateral USD currency return ($\Delta q_{c,t}$), the country index return, and the world index return. All columns include country-industry fixed effects, and even-numbered columns add country $\times$ year fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. Using export/import-weighted returns								
	Value-weighted returns				Equally weighted returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^X$	-0.092 (-2.60)	-0.095 (-2.54)	-0.061 (-2.34)	-0.073 (-2.80)	-0.073 (-1.93)	-0.086 (-2.06)	-0.043 (-1.56)	-0.065 (-2.14)
$\Delta q_{c,i,t}^M$	0.130 (3.16)	0.135 (2.96)	0.089 (2.70)	0.100 (2.99)	0.148 (3.56)	0.154 (3.29)	0.109 (3.18)	0.122 (3.38)
$R_{c,t}$			0.623 (15.62)	0.611 (15.89)			0.591 (16.03)	0.578 (15.97)
$R_{WLD,t}$	0.808 (15.01)	0.781 (16.08)	0.273 (6.28)	0.263 (6.50)	0.772 (13.08)	0.738 (13.79)	0.265 (5.60)	0.249 (5.56)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.128	0.149	0.201	0.213	0.117	0.145	0.181	0.201
N	313,610	313,610	313,610	313,610	313,610	313,610	313,610	313,610

Panel B. Using export-to-import currency returns								
	Value-weighted returns				Equally weighted returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^{XM}$	0.098 (2.78)	0.101 (2.69)	0.066 (2.39)	0.079 (2.85)	0.087 (2.27)	0.099 (2.40)	0.057 (1.83)	0.078 (2.43)
$\Delta q_{c,t}$	-0.009 (-0.28)	-0.013 (-0.39)	0.021 (0.99)	0.015 (0.75)	0.030 (0.87)	0.021 (0.64)	0.058 (2.27)	0.047 (2.05)
$R_{c,t}$			0.624 (15.57)	0.612 (15.84)			0.592 (15.96)	0.579 (15.90)
$R_{WLD,t}$	0.812 (14.96)	0.786 (15.80)	0.269 (6.03)	0.260 (6.22)	0.769 (12.89)	0.737 (13.31)	0.253 (5.18)	0.240 (5.09)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.128	0.149	0.201	0.213	0.117	0.144	0.181	0.201
N	313,610	313,610	313,610	313,610	313,610	313,610	313,610	313,610

Table 3: Trade intensity and country-industry currency risk exposure

This table reports the results of interacting the trade-weighted currency returns with the country-industry trade share, defined as the ratio of exports plus imports to output plus intermediate inputs. The dependent variables are country-industry monthly equity returns. Columns (1)–(4) report value-weighted returns and columns (5)–(8) report equally weighted returns. In Panel A, the independent variables include the export-weighted ($\Delta q_{c,i,t}^X$) and import-weighted ($\Delta q_{c,i,t}^M$) currency returns, both individually and interacted with the country-industry trade share ($\%Trd_{c,i}$), as well as the country index return ($R_{c,t}$) and the world index return ($R_{WLD,t}$). The standalone $\%Trd_{c,i}$ is absorbed by country $\times$ industry fixed effects. In Panel B, the independent variables are the export-to-import currency return ($\Delta q_{c,i,t}^{XM}$), its interaction with $\%Trd_{c,i}$, the bilateral USD currency return ($\Delta q_{c,t}$), the country index return, and the world index return. All columns include country-industry fixed effects, and even-numbered columns add country $\times$ year fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. Using export/import-weighted returns								
	Value-weighted returns				Equally weighted returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^X \times \%Trd_{c,i}$	-0.849 (-2.80)	-0.741 (-2.46)	-0.563 (-2.82)	-0.501 (-2.54)	-0.767 (-2.83)	-0.653 (-2.52)	-0.496 (-2.68)	-0.427 (-2.41)
$\Delta q_{c,i,t}^M \times \%Trd_{c,i}$	1.281 (4.26)	1.176 (3.90)	0.707 (3.48)	0.673 (3.36)	1.340 (5.04)	1.243 (4.84)	0.796 (4.16)	0.769 (4.24)
$\Delta q_{c,i,t}^X$	0.060 (1.04)	0.042 (0.69)	0.039 (1.41)	0.019 (0.64)	0.065 (1.20)	0.036 (0.64)	0.046 (1.69)	0.015 (0.47)
$\Delta q_{c,i,t}^M$	-0.107 (-1.74)	-0.089 (-1.34)	-0.040 (-1.03)	-0.027 (-0.67)	-0.103 (-1.81)	-0.084 (-1.38)	-0.039 (-1.06)	-0.025 (-0.63)
$R_{c,t}$			0.623 (15.59)	0.611 (15.87)			0.591 (15.98)	0.577 (15.93)
$R_{WLD,t}$	0.807 (14.99)	0.780 (16.05)	0.273 (6.29)	0.263 (6.50)	0.771 (13.06)	0.737 (13.75)	0.265 (5.60)	0.249 (5.56)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.128	0.150	0.201	0.213	0.117	0.145	0.181	0.201
N	313,610	313,610	313,610	313,610	313,610	313,610	313,610	313,610
Panel B. Using export-to-import currency returns								
	Value-weighted returns				Equally weighted returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^{XM} \times \%Trd_{c,i}$	0.949 (2.96)	0.840 (2.62)	0.619 (3.06)	0.557 (2.78)	0.938 (3.20)	0.817 (2.88)	0.624 (3.30)	0.548 (3.05)
$\Delta q_{c,i,t}^{XM}$	-0.070 (-1.17)	-0.052 (-0.83)	-0.043 (-1.54)	-0.022 (-0.75)	-0.078 (-1.43)	-0.049 (-0.85)	-0.053 (-2.01)	-0.021 (-0.70)
$\Delta q_{c,t}$	-0.007 (-0.21)	-0.011 (-0.33)	0.022 (1.07)	0.016 (0.82)	0.032 (0.94)	0.023 (0.70)	0.060 (2.33)	0.048 (2.11)
$R_{c,t}$			0.623 (15.56)	0.611 (15.83)			0.592 (15.95)	0.578 (15.89)
$R_{WLD,t}$	0.812 (14.99)	0.786 (15.82)	0.269 (6.04)	0.261 (6.23)	0.769 (12.91)	0.737 (13.33)	0.254 (5.18)	0.240 (5.10)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.128	0.149	0.201	0.213	0.117	0.145	0.181	0.201
N	313,610	313,610	313,610	313,610	313,610	313,610	313,610	313,610

Table 4: Currency returns and country-industry operating outcomes

This table reports the relationship between trade-weighted currency returns and changes in country-industry profit margins, revenue, and cost of goods sold (COGS), at annual ($h = 0$) and four-year cumulative ($h = 3$) horizons. In Panel A the dependent variable is the change in gross profit margin, and the independent variables include the import- and export-weighted currency returns, both contemporaneous and one-period lagged. In Panel B, the dependent variable is the log change in revenue, and in Panel C it is the log change in COGS, all measured at the country-industry level after summing firm-level revenue and COGS. All specifications control for the one-period lagged outcome, the contemporaneous and lagged global aggregate outcome, and country-industry fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. Profit margin

	$\Delta \text{Profit margin}_t$				$\sum_{k=0}^3 \Delta \text{Profit margin}_{t+k}$			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^X$	-0.010 (-1.46)	-0.013 (-1.75)			-0.041 (-1.86)	-0.042 (-2.04)		
$\Delta q_{c,i,t}^M$	-0.000 (-0.03)	0.003 (0.42)			0.035 (1.78)	0.038 (2.06)		
$\Delta q_{c,i,t-1}^X$		-0.015 (-3.23)				-0.037 (-2.77)		
$\Delta q_{c,i,t-1}^M$		0.030 (4.89)				0.047 (3.13)		
$\Delta q_{c,i,t}^{XM}$			0.008 (1.18)	0.011 (1.37)			0.035 (1.86)	0.037 (2.02)
$\Delta q_{c,i,t-1}^{XM}$				0.018 (4.04)				0.039 (2.86)
$\Delta q_{c,t}$			-0.002 (-0.54)	-0.003 (-0.91)			-0.034 (-3.11)	-0.033 (-3.25)
$\Delta q_{c,t-1}$				0.003 (0.66)				-0.014 (-1.49)
Profit margin controls	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.146	0.151	0.146	0.150	0.416	0.423	0.417	0.424
N	23,488	22,940	23,488	22,940	19,945	19,448	19,945	19,448

Panel B. Revenue

	$\Delta \ln \text{Rev}_t$		$\sum_{k=0}^3 \Delta \ln \text{Rev}_{t+k}$	
	(1)	(2)	(3)	(4)
$\Delta q_{c,i,t}^X$	-0.020 (-0.27)	-0.031 (-0.40)	-0.177 (-2.26)	-0.179 (-2.96)
$\Delta q_{c,i,t-1}^X$		-0.025 (-0.76)		-0.328 (-3.53)
$\Delta q_{c,t}$	-0.204 (-2.38)	-0.199 (-2.38)	0.061 (0.56)	0.056 (0.65)
$\Delta q_{c,t-1}$		-0.101 (-1.70)		0.348 (4.08)
Rev controls	Y	Y	Y	Y
Ctr $\times$ Ind FE	Y	Y	Y	Y
R ²	0.244	0.248	0.372	0.383
N	23,488	22,940	19,945	19,448

Panel C. Cost of goods sold

	$\Delta \ln \text{COGS}_t$		$\sum_{k=0}^3 \Delta \ln \text{COGS}_{t+k}$	
	(1)	(2)	(3)	(4)
$\Delta q_{c,i,t}^M$	-0.101 (-1.04)	-0.128 (-1.27)	-0.303 (-3.05)	-0.291 (-3.50)
$\Delta q_{c,i,t-1}^M$		-0.127 (-2.01)		-0.373 (-5.00)
$\Delta q_{c,t}$	-0.153 (-1.64)	-0.142 (-1.53)	0.179 (1.32)	0.154 (1.37)
$\Delta q_{c,t-1}$		-0.044 (-0.56)		0.383 (4.62)
COGS controls	Y	Y	Y	Y
Ctr $\times$ Ind FE	Y	Y	Y	Y
R ²	0.233	0.241	0.354	0.367
N	23,488	22,940	19,945	19,448

Table 5: Currency invoicing and country-industry revenue and cost growth

This table describes the result of the panel regressions, where country-industry log revenue (Panel A) and log cost of goods sold (Panel B, COGS) growth is regressed on export- and import-weighted currency returns and their interaction to the share of trade invoiced in US dollars, at one-year ($h = 0$) and four-year cumulative ($h = 3$) horizons. In Panel A the independent variables include the export-weighted currency return interacted with the USD export-invoicing share ($INV_{c,t}^X$), both contemporaneous and one-year lagged, along with USD currency returns and the invoicing share. In Panel B the import-weighted currency return interacted with the USD import-invoicing share ($INV_{c,t}^M$). The sample sizes differ across panels because of differences in international data availability for export and import invoicing shares. The “ $\sum$ interactions” row reports the sum of the contemporaneous and lagged interaction coefficients. All specifications control for the listed growth controls and country-industry fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. Revenue					Panel B. Cost of goods sold				
	1Y growth		4Y growth			1Y growth		4Y growth	
	(1)	(2)	(3)	(4)		(1)	(2)	(3)	(4)
$\Delta q_{c,i,t}^X \times INV_{c,t}^X$	0.151 (1.78)	0.128 (1.72)	0.067 (0.51)	0.104 (0.82)	$\Delta q_{c,i,t}^M \times INV_{c,t}^M$	0.321 (2.48)	0.359 (3.03)	-0.025 (-0.13)	0.005 (0.03)
$\Delta q_{c,i,t-1}^X \times INV_{c,t}^X$		0.211 (2.09)		-0.086 (-0.72)	$\Delta q_{c,i,t-1}^M \times INV_{c,t}^M$		0.160 (0.90)		-0.240 (-1.36)
$\Delta q_{c,i,t}^X$	-0.082 (-1.30)	-0.074 (-1.11)	-0.183 (-2.35)	-0.217 (-2.54)	$\Delta q_{c,i,t}^M$	-0.296 (-2.48)	-0.348 (-3.07)	-0.255 (-2.24)	-0.275 (-2.24)
$\Delta q_{c,i,t-1}^X$		-0.110 (-1.61)		-0.218 (-1.73)	$\Delta q_{c,i,t-1}^M$		-0.216 (-2.13)		-0.168 (-1.11)
$\Delta q_{c,t}^X$	-0.230 (-2.47)	-0.225 (-2.56)	0.038 (0.37)	0.042 (0.56)	$\Delta q_{c,t}^M$	-0.154 (-1.69)	-0.141 (-1.61)	0.168 (1.34)	0.154 (1.51)
$\Delta q_{c,t-1}^X$		-0.133 (-1.95)		0.318 (3.66)	$\Delta q_{c,t-1}^M$		-0.066 (-0.91)		0.340 (3.85)
$INV_{c,t}^X$	-0.028 (-0.44)	-0.088 (-1.40)	-0.402 (-3.39)	-0.340 (-2.69)	$INV_{c,t}^M$	-0.036 (-1.00)	-0.045 (-1.24)	-0.097 (-1.60)	-0.089 (-1.51)
$\sum$ interactions		0.339 (3.14)		0.018 (0.08)	$\sum$ interactions		0.519 (2.58)		-0.235 (-0.75)
Rev controls	Y	Y	Y	Y	COGS controls	Y	Y	Y	Y
Ctr $\times$ Ind FE	Y	Y	Y	Y	Ctr $\times$ Ind FE	Y	Y	Y	Y
R ²	0.245	0.248	0.376	0.386	R ²	0.237	0.246	0.353	0.364
N	20,791	20,282	17,647	17,187	N	18,936	18,485	16,054	15,645

Table 6: Country index returns and exposure to trade-weighted currency returns

This table describes the country-level results where MSCI country index returns are regressed on trade-weighted currency returns computed at the country level. Columns (1)–(2) and (5)–(6) use the export- and import-weighted currency returns $\Delta q_{c,t}^X$ and $\Delta q_{c,t}^M$ separately, and columns (3)–(4) and (7)–(8) report the export-to-import differential $\Delta q_{c,t}^{XM}$ alongside the bilateral USD return $\Delta q_{c,t}$. The dependent variable in columns (1)–(4) is the raw country index return, and in columns (5)–(8) it is the return in excess of the US equity return. The raw return specifications control for world equity returns, while excess-return specifications do not. Odd-numbered columns include country fixed effects, while even-numbered columns add country $\times$ year fixed effects. Panel A uses weights computed from OECD Input-Output data, and Panel B uses IMF DOTS trade flows. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. OECD Input-Output sample								
	Country index returns				Country excess-to-US returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}^X$	-0.844 (-2.89)	-0.808 (-2.87)			-0.701 (-2.53)	-0.683 (-2.68)		
$\Delta q_{c,t}^M$	0.865 (2.92)	0.836 (2.91)			0.719 (2.56)	0.702 (2.70)		
$\Delta q_{c,t}^{XM}$			0.812 (2.76)	0.783 (2.75)			0.715 (2.44)	0.683 (2.57)
$\Delta q_{c,t}$			-0.036 (-1.01)	-0.032 (-0.82)			0.008 (0.19)	-0.004 (-0.10)
$R_{WLD,t}$	0.870 (24.11)	0.856 (26.24)	0.881 (23.30)	0.866 (24.32)				
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.412	0.468	0.413	0.468	0.008	0.104	0.008	0.104
N	7,653	7,650	7,653	7,650	7,327	7,324	7,327	7,324
Panel B. IMF DOTS sample								
	Country index returns				Country excess-to-US returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}^X$	-1.039 (-4.07)	-0.955 (-3.76)			-0.784 (-2.86)	-0.705 (-2.53)		
$\Delta q_{c,t}^M$	1.085 (4.18)	1.006 (3.85)			0.823 (2.98)	0.744 (2.65)		
$\Delta q_{c,t}^{XM}$			1.007 (3.82)	0.928 (3.53)			0.796 (2.83)	0.702 (2.49)
$\Delta q_{c,t}$			-0.034 (-0.95)	-0.030 (-0.76)			-0.001 (-0.02)	-0.011 (-0.29)
$R_{WLD,t}$	0.854 (25.54)	0.841 (27.39)	0.867 (24.45)	0.854 (25.25)				
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.410	0.466	0.410	0.466	0.009	0.104	0.009	0.103
N	6,975	6,972	6,975	6,972	6,649	6,646	6,649	6,646

Table 7: Net export and country-level currency risk exposure

This table reports regressions of country index returns on the USD currency return $\Delta q_{c,t}$ interacted with measures of country's aggregate trade position. The first four columns use OECD Input-Output trade weights and the remainder use IMF DOTS trade flows. Within each sample, the first two columns use raw country index returns, while the last two use country index returns in excess of the US equity return. Odd-numbered columns interact USD returns with the lagged normalized net export $NX_{c,t}^- = (EXP_{c,t}^- - IMP_{c,t}^-)/(EXP_{c,t}^- + IMP_{c,t}^-)$. Even-numbered columns instead interact USD returns separately with the lagged export share $\%EXP_{c,t}^-$ and import share $\%IMP_{c,t}^-$, whose denominator is $IN + OUT$ in the OECD Input-Output sample and GDP in the IMF DOTS sample. All regressions include country $\times$ year fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

	OECD Input-Output sample				IMF DOTS sample			
	Raw returns		Excess-to-US returns		Raw returns		Excess-to-US returns	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}$	0.061 (0.96)	-0.471 (-3.15)	0.089 (1.36)	-0.267 (-1.98)	-0.073 (-1.84)	-0.145 (-2.48)	-0.047 (-1.16)	-0.091 (-1.54)
$\Delta q_{c,t} \times NX_{c,t}^-$	-0.658 (-2.50)		-0.638 (-2.31)		-0.296 (-1.43)		-0.289 (-1.37)	
$\Delta q_{c,t} \times \%EXP_{c,t}^-$		-0.095 (-0.39)		-0.268 (-1.02)		-0.618 (-1.07)		-0.655 (-1.13)
$\Delta q_{c,t} \times \%IMP_{c,t}^-$		1.244 (3.13)		1.032 (2.60)		0.867 (1.67)		0.794 (1.50)
$R_{WLD,t}$	0.868 (24.38)	0.869 (24.32)			0.859 (24.90)	0.861 (25.30)		
Country $\times$ Year FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.468	0.469	0.104	0.104	0.464	0.466	0.102	0.104
N	7,650	7,650	7,324	7,324	6,972	6,672	6,646	6,346

Table 8: Gross profit margin and USD currency beta

This table describes the results of the panel interactive regressions, where country index returns are regressed on the bilateral USD currency return $\Delta q_{c,t}$ interacted with the aggregate profit margin $PM_{c,t}^-$ and with additional country characteristics. Panel A uses the OECD Input-Output sample and Panel B uses IMF DOTS. Columns (1)–(4) of each panel use raw country index returns and control for the world index return $R_{WLD,t}$, while columns (5)–(8) use country index returns in excess of the US equity return. For each specification, the first column reports the interactive profit margin effect, while the remaining columns iteratively introduce interactions with additional country characteristics (the price-to-sales ratio $PS_{c,t}^-$, $NX_{c,t}^-$, $NIIP_{c,t}^-$) and financial controls ($CorpDebt_{c,t}^-$, $GrossLib_{c,t}^-$, and $NetFX_{c,t}^-$). All regressions include country $\times$ year fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. OECD Input-Output sample								
	Raw country index returns				Country excess-to-US returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}$	0.418 (5.08)	0.415 (5.02)	0.306 (3.21)	0.337 (3.00)	0.352 (4.06)	0.341 (3.84)	0.226 (2.25)	0.331 (2.71)
$\Delta q_{c,t} \times PM_{c,t}^-$	-1.363 (-7.37)	-1.356 (-7.15)	-1.269 (-5.80)	-1.236 (-5.16)	-1.097 (-5.16)	-1.068 (-4.83)	-0.914 (-3.65)	-0.981 (-3.64)
$\Delta q_{c,t} \times PS_{c,t}^-$		-0.007 (-0.16)	-0.012 (-0.26)	0.044 (0.86)		-0.030 (-0.62)	-0.029 (-0.58)	0.042 (0.78)
$\Delta q_{c,t} \times NX_{c,t}^-$			0.167 (0.56)	0.437 (1.19)			0.057 (0.19)	0.360 (0.96)
$\Delta q_{c,t} \times NIIP_{c,t}^-$			-0.020 (-2.34)	-0.022 (-1.37)			-0.024 (-2.92)	-0.019 (-1.17)
$\Delta q_{c,t} \times CorpDebt_{c,t}^-$				-2.448 (-2.02)				-2.898 (-2.53)
$\Delta q_{c,t} \times GrossLib_{c,t}^-$				-0.010 (-0.16)				0.021 (0.30)
$\Delta q_{c,t} \times NetFX_{c,t}^-$				-0.091 (-1.43)				-0.092 (-1.36)
Country $\times$ Year FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.473	0.473	0.488	0.478	0.110	0.110	0.116	0.112
N	7,650	7,650	7,141	5,728	6,989	6,989	6,504	5,200

Panel B. IMF DOTS sample								
	Raw country index returns				Country excess-to-US returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}$	0.436 (5.21)	0.433 (5.13)	0.323 (3.81)	0.375 (3.40)	0.360 (4.08)	0.348 (3.85)	0.230 (2.50)	0.356 (3.00)
$\Delta q_{c,t} \times PM_{c,t}^-$	-1.426 (-7.28)	-1.419 (-7.04)	-1.221 (-6.03)	-0.996 (-4.43)	-1.143 (-5.12)	-1.112 (-4.80)	-0.891 (-3.76)	-0.747 (-2.87)
$\Delta q_{c,t} \times PS_{c,t}^-$		-0.008 (-0.18)	-0.007 (-0.16)	0.068 (1.27)		-0.032 (-0.65)	-0.027 (-0.55)	0.065 (1.16)
$\Delta q_{c,t} \times NX_{c,t}^-$			0.099 (0.46)	0.055 (0.19)			0.077 (0.35)	-0.003 (-0.01)
$\Delta q_{c,t} \times NIIP_{c,t}^-$			-0.022 (-2.24)	-0.016 (-0.85)			-0.025 (-2.79)	-0.012 (-0.66)
$\Delta q_{c,t} \times CorpDebt_{c,t}^-$				-1.482 (-1.27)				-1.963 (-1.78)
$\Delta q_{c,t} \times GrossLib_{c,t}^-$				-0.049 (-0.70)				-0.025 (-0.36)
$\Delta q_{c,t} \times NetFX_{c,t}^-$				-0.162 (-2.24)				-0.162 (-2.20)
Country $\times$ Year FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.472	0.472	0.485	0.475	0.109	0.109	0.116	0.115
N	6,972	6,972	6,559	5,264	6,646	6,646	6,233	4,988

Table 9: Currency beta and gross profit margin: Cross-section vs time-series

This table summarizes the results of regressions after separating the profit-margin interaction into its cross-sectional and time-series components, testing whether a unit's currency exposure varies with its own profit margin over time. Panel A regresses country index returns on $\Delta q_{c,t}$ and its interaction with the profit margin $PM_{c,t}^-$, controlling for the world index return, with country $\times$ year fixed effects. Columns (1)–(3) of the panel use OECD Input-Output weights and (4)–(6) use IMF DOTS. Panel B shows the result at the country-industry level with country $\times$ year fixed effects. The first three columns are for value-weighted returns and the remainder for equally weighted returns. In columns (3) and (6), each unit (a country in Panel A and a country-industry in Panel B) has its own time-invariant currency slope. Panel B columns (3) and (6) additionally include country-industry fixed effects. The between-unit margin interaction is absorbed by these slopes. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. Using country index returns as the dependent variable						
	OECD Input-Output sample			IMF DOTS sample		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta q_{c,t}$	0.418 (5.08)	0.636 (5.87)		0.436 (5.21)	0.627 (5.86)	
$\Delta q_{c,t} \times PM_{c,t}^-$	-1.363 (-7.37)			-1.426 (-7.28)		
$\Delta q_{c,t} \times \widetilde{PM}_{c,t}^-$		-0.535 (-2.05)	-0.350 (-1.28)		-0.686 (-2.38)	-0.475 (-1.57)
$\Delta q_{c,t} \times \overline{PM}_c^-$		-1.992 (-7.75)			-1.974 (-7.78)	
$R_{WLD,t}$	0.856 (24.08)	0.863 (23.93)	0.852 (23.64)	0.847 (24.78)	0.855 (24.60)	0.846 (24.46)
Country $\times$ Year FE	Y	Y	Y	Y	Y	Y
Country-specific FX slope	N	N	Y	N	N	Y
R ²	0.473	0.476	0.498	0.472	0.474	0.498
N	7,650	7,650	7,650	6,972	6,972	6,972
Panel B. Using country-industry returns as the dependent variable						
	Value-weighted returns			Equally weighted returns		
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta q_{c,t}$	0.141 (3.24)	0.200 (4.15)		0.172 (3.74)	0.224 (4.61)	
$PM_{c,i,t}^-$	0.004 (2.63)			0.002 (1.17)		
$\Delta q_{c,t} \times PM_{c,i,t}^-$	-0.447 (-7.02)			-0.427 (-6.18)		
$\Delta q_{c,t} \times \widetilde{PM}_{c,i,t}^-$		-0.290 (-2.83)	-0.294 (-3.07)		-0.286 (-2.54)	-0.291 (-2.90)
$\Delta q_{c,t} \times \overline{PM}_{c,i}^-$		-0.588 (-6.66)			-0.552 (-6.18)	
$R_{WLD,t}$	0.756 (14.96)	0.758 (14.93)	0.747 (14.75)	0.698 (12.52)	0.700 (12.49)	0.688 (12.32)
Country $\times$ Year FE	Y	Y	Y	Y	Y	Y
Country-industry-specific FX slope	N	N	Y	N	N	Y
R ²	0.123	0.123	0.142	0.118	0.118	0.134
N	342,739	342,739	342,739	342,739	342,739	342,739

Table 10: Currency risk, profit margin, and trade intensity: Triple interaction

This table describes the result of the regression, where country index returns are regressed on the bilateral USD currency return $\Delta q_{c,t}$ and its interactions with the aggregate profit margin $PM_{c,t}^-$, trade intensity $\%Trd_{c,t}^-$, and their triple interaction, where $\%Trd_{c,t}^-$ is defined as the ratio of exports plus imports to output plus inputs in the OECD Input-Output sample and to GDP in the IMF DOTS sample. Panel A uses the OECD Input-Output sample and Panel B uses IMF DOTS flows. In each panel, columns (1)–(4) report raw country index returns and control for the world index return $R_{WLD,t}$, and columns (5)–(8) report the results for country index returns in excess of the US equity return. All regressions include country $\times$ year fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. OECD Input-Output sample								
	Raw country index returns				Country excess-to-US returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}$	0.418 (5.08)	-0.454 (-3.03)	0.238 (1.44)	-0.573 (-1.80)	0.352 (4.06)	-0.251 (-1.86)	0.314 (1.85)	-0.535 (-1.81)
$\Delta q_{c,t} \times PM_{c,t}^-$	-1.363 (-7.37)		-1.319 (-6.89)	0.689 (1.21)	-1.097 (-5.16)		-1.088 (-4.98)	1.017 (1.83)
$\Delta q_{c,t} \times \%Trd_{c,t}^-$		0.448 (2.79)	0.180 (1.29)	1.128 (3.10)		0.259 (1.70)	0.038 (0.26)	1.031 (2.95)
$\Delta q_{c,t} \times PM_{c,t}^- \times \%Trd_{c,t}^-$				-2.416 (-3.38)				-2.533 (-3.42)
$R_{WLD,t}$	0.856 (24.08)	0.867 (24.04)	0.856 (23.98)	0.857 (24.01)				
Country $\times$ Year FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.473	0.468	0.474	0.475	0.110	0.103	0.110	0.112
N	7,650	7,650	7,650	7,650	6,989	6,989	6,989	6,989

Panel B. IMF DOTS sample								
	Raw country index returns				Country excess-to-US returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}$	0.436 (5.21)	-0.130 (-2.18)	0.399 (3.89)	0.157 (1.16)	0.360 (4.08)	-0.077 (-1.28)	0.344 (3.34)	0.071 (0.52)
$\Delta q_{c,t} \times PM_{c,t}^-$	-1.426 (-7.28)		-1.376 (-6.85)	-0.546 (-1.65)	-1.143 (-5.12)		-1.104 (-4.91)	-0.169 (-0.50)
$\Delta q_{c,t} \times \%Trd_{c,t}^-$		0.151 (1.46)	0.020 (0.21)	0.648 (2.88)		0.095 (0.89)	-0.012 (-0.12)	0.694 (3.03)
$\Delta q_{c,t} \times PM_{c,t}^- \times \%Trd_{c,t}^-$				-2.179 (-3.57)				-2.451 (-4.01)
$R_{WLD,t}$	0.847 (24.78)	0.861 (25.21)	0.852 (25.24)	0.852 (25.28)				
Country $\times$ Year FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.472	0.466	0.473	0.474	0.109	0.104	0.111	0.113
N	6,972	6,672	6,672	6,672	6,646	6,346	6,346	6,346

A Model appendix

This appendix first derives the profit response to the full vector of partner-specific bilateral exchange rates and shows how the responses depend on export- and import-weighted currency returns. Then, a special case is considered with a CES demand function with one export destination and one import origin. The model provides a closed-form solution. The final subsection introduces dollar invoicing and derives the short-run responses of revenue and cost.

A.1 The envelope result

Let d be a trade partner country of the focal firm, where q_d denotes the units of partner- d currency per unit of home currency. Let $p_d X_d$ denote export sales to d and $w_d M_d$ imported-input spending from d , both in partner d 's currency. Domestic revenue S^D and cost C^D are in home currency, and $\mathbf{z}$ is the collection of the firm's choice variables.

Define export revenue and imported-input spending in the home currency as

$$S_d^X \equiv \frac{p_d X_d}{q_d}, \quad C_d^M \equiv \frac{w_d M_d}{q_d}. \quad (\text{A.1})$$

The operating profit in the home currency is

$$\Pi(\mathbf{z}, \mathbf{q}) = S^D + \sum_d S_d^X - C^D - \sum_d C_d^M. \quad (\text{A.2})$$

The sums include the union of all export destinations and import origins, with the corresponding term set to zero when the firm does not export to or import from the partner.

Let $\mathbf{z}^*(\mathbf{q})$ denote the firm's profit-maximizing choices and $\Pi^*(\mathbf{q}) \equiv \Pi(\mathbf{z}^*(\mathbf{q}), \mathbf{q})$. For each partner d , the chain rule gives

$$\frac{d\Pi^*}{d \ln q_d} = \frac{\partial \Pi}{\partial \ln q_d} + \nabla_{\mathbf{z}} \Pi \frac{\partial \mathbf{z}^*}{\partial \ln q_d}. \quad (\text{A.3})$$

At an interior optimum, with exchange rates entering profit only through currency conversion, $\nabla_{\mathbf{z}} \Pi = 0$ and the envelope theorem reduces the response to its direct effect. Holding the firm's choices fixed,

$$\frac{\partial \Pi}{\partial \ln q_d} = \frac{\partial S_d^X}{\partial \ln q_d} - \frac{\partial C_d^M}{\partial \ln q_d} = -S_d^X + C_d^M. \quad (\text{A.4})$$

Combining the last two equations yields the partner-level envelope result that the main text requires.

A.2 A model with CES demand

A.2.1 Setup

For a closed-form example, specialize to one export destination, denoted d_X , and one import origin, denoted d_M . Write $q^X \equiv q_{d_X}$ and $q^M \equiv q_{d_M}$.⁶

Production uses one unit of an input composite per unit of output. A share κ is imported at price w in the supplier's currency, and the remaining share is sourced domestically at price w_D . Marginal cost in the firm's currency is

$$c(q^M) = (1 - \kappa)w_D + \kappa \frac{w}{q^M}. \quad (\text{A.5})$$

⁶The empirical analysis uses nominal exchange-rate returns. At a monthly frequency, nominal and real exchange-rate changes are assumed to track each other closely because price levels adjust much more slowly than exchange rates.

The firm generates revenue from two markets. At home, it faces CES demand with elasticity $\sigma > 1$ (Dixit and Stiglitz, 1977),

$$X_D = A_D p_D^{-\sigma}, \quad (\text{A.6})$$

where p_D is the price in the firm's own currency. Abroad, it faces the same type of demand curve,

$$X_F = A_F p_F^{-\sigma}, \quad (\text{A.7})$$

where p_F is set in the destination's currency. The firm receives p_F/q^X in its home currency for each unit sold abroad.

A.2.2 The long-run benchmark

The firm's optimal prices are constant markups over marginal cost. In the domestic market,

$$p_D^* = \mu c, \quad (\text{A.8})$$

where $\mu \equiv \frac{\sigma}{\sigma - 1} > 1$. In the foreign market, maximizing $(p_F/q^X - c)A_F p_F^{-\sigma}$ over p_F implies

$$p_F^* = \mu c q^X. \quad (\text{A.9})$$

The export price is quoted in the destination's currency and moves one-for-one with q^X . The firm therefore receives the same constant markup in its own currency in both markets. An appreciation raises the price faced by foreign buyers and lowers export quantity.

Substituting the two optimal prices back into demand gives total revenue,

$$S^* = \mu^{1-\sigma} c^{1-\sigma} \left[A_D + A_F (q^X)^{-\sigma} \right], \quad (\text{A.10})$$

and profit,

$$\Pi^* = (\mu - 1) \mu^{-\sigma} c^{1-\sigma} [A_D + A_F (q^X)^{-\sigma}]. \quad (\text{A.11})$$

Note that the profit margin is the ratio of the two,

$$\frac{\Pi^*}{S^*} = \frac{\mu - 1}{\mu} = \sigma^{-1}, \quad (\text{A.12})$$

which is the Lerner index (Lerner, 1934). The foreign share of revenue and the imported share of marginal cost are, respectively,

$$\theta^X = \frac{A_F (q^X)^{-\sigma}}{A_D + A_F (q^X)^{-\sigma}} \quad \text{and} \quad \theta^M = \frac{\kappa w / q^M}{c}. \quad (\text{A.13})$$

Using the envelope condition and differentiating optimized profit with respect to the export rate, holding the import rate fixed gives

$$\frac{\partial \Pi^*}{\partial \ln q^X} = -\mu^{1-\sigma} c^{1-\sigma} A_F (q^X)^{-\sigma} = -S^{X*}. \quad (\text{A.14})$$

For the import rate, holding the export rate fixed, it follows that

$$\frac{\partial \Pi^*}{\partial \ln q^M} = (\sigma - 1) \theta^M \Pi^* = \theta^M C^* = C^{M*}, \quad (\text{A.15})$$

where the second equality follows from constant market pricing rule.

A.2.3 The short run effect of dollar invoicing

In practice, a large share of world trade is invoiced in a small number of currencies, most notably the US dollar (Gopinath, Boz, Casas, Díez, Gourinchas, and Plagborg-Møller, 2020; Boz, Casas, Georgiadis, Gopinath, Le Mezo, Mehl, and Nguyen, 2022). Suppose that invoicing on both the cost and revenue sides is a mix of Producer Currency Pricing (PCP, share $1 - \alpha$) and Dominant Currency Pricing (DCP, share α).

Unlike the flexible-price long-run benchmark, the short-run analysis concerns contracts written in nominal currency units. Let q denote the value of the firm's currency against the dollar. Let P denote the home price level, normalized to $P = 1$ at the contracting date and held fixed until the contract resets. Nominal and real exchange rates therefore move one-for-one during the contract period. Accordingly, q^X , q^M , and q can be read as either nominal or real exchange rates, and all short-run expressions are in common real home-currency units.

Invoice prices equal the long-run optimum when the contract is written, but remain fixed in the currency specified by the contract until the next reset. The imported input is priced in the seller's currency with probability $1 - \alpha$ and in dollars with probability α . Short-run unit cost is

$$c^{SR} = (1 - \kappa) w_D + \kappa \left[(1 - \alpha) \frac{\bar{w}}{q^M} + \alpha \frac{\bar{w}_\$}{q} \right], \quad (\text{A.16})$$

where $\bar{w}$ is the preset foreign-currency price under PCP and $\bar{w}_\text{\$}$ is the corresponding price fixed in dollars at the contracting date. Under the normalization, $\bar{w}/q^M$ and $\bar{w}_\text{\$}/q$ are directly comparable real home-currency costs per unit of the imported input. The first depends on the bilateral rate against the supplier and the second on the dollar rate.

Under PCP, the invoice of the product sold is set at home-currency at $\bar{p}_F = \mu\bar{c}$, so the buyer faces $\bar{p}_F q^X$ in its own currency and buys $A_F(\bar{p}_F q^X)^{-\sigma}$. Under DCP, the firm collects $\bar{p}_\text{\$}/q$ per unit in its own currency, while the buyer pays $\bar{p}_\text{\$} q^X/q$ in the buyer's currency. The quantity sold under DCP is therefore $A_F(\bar{p}_\text{\$} q^X/q)^{-\sigma}$. At the contracting date the invoice is set at the long-run optimum, $\bar{p}_F = \bar{p}_\text{\$} = \mu\bar{c}$.

Total output is

$$X^{SR} = (\mu\bar{c})^{-\sigma} \left[A_D + A_F (q^X)^{-\sigma} ((1 - \alpha) + \alpha q^\sigma) \right], \quad (\text{A.17})$$

The domestic and PCP lines each collect $\mu\bar{c}$ per unit in home currency, while the DCP line collects $\bar{p}_\$/q = \mu\bar{c}/q$. So the short-run revenue is $\mu\bar{c} X_D + \mu\bar{c}(1 - \alpha)X_F^{PCP} + (\mu\bar{c}/q)\alpha X_F^{DCP}$ or

$$S^{SR} = (\mu\bar{c})^{1-\sigma} \left[A_D + A_F (q^X)^{-\sigma} ((1 - \alpha) + \alpha q^{\sigma-1}) \right]. \quad (\text{A.18})$$

Every unit costs c^{SR} from Equation (A.16), so short-run profit is

$$\Pi^{SR} = S^{SR} - c^{SR} X^{SR}. \quad (\text{A.19})$$

Two extreme cases make the point cleanly. Under full PCP ($\alpha = 0$), every invoice is written in the seller's currency, so the home-currency cost of imported inputs moves with q^M and the buyer's price moves with q^X , as in the flexible-price benchmark. The short-run problem is not identical to the long-run one, however, because output prices stay at $\mu\bar{c}$ rather than repricing to μc^{SR} , so an import-rate shock changes unit cost without changing output. Under full DCP ($\alpha = 1$), the cost side depends only on the firm's own dollar rate q , so the bilateral rate against the supplier does not affect profit. The revenue side remains exposed to the bilateral rate because export quantity responds to the price faced by the foreign buyer.

The general case follows by direct differentiation of Equations (A.16) and (A.18). In the short run prices are preset, so holding the dollar rate fixed,

$$\frac{\partial \ln c^{SR}}{\partial \ln q^M} = -\theta_{SR}^{M,P}, \quad \frac{\partial \ln S^{SR}}{\partial \ln q^X} = -\sigma \theta_{SR}^X, \quad (\text{A.20})$$

where $\theta_{SR}^{M,P} = \kappa(1-\alpha)(\bar{w}/q^M)/c^{SR}$ is the share of short-run unit cost invoiced in the supplier's currency, and θ_{SR}^X is the foreign share of S^{SR} .

The first elasticity falls to zero as $\alpha \rightarrow 1$, so dollar invoicing weakens the influence of the bilateral import rate. Evaluated at the contracting date, where $q = 1$, the second is the same for every α , so dollar invoicing alone does not attenuate the bilateral revenue elasticity, and such attenuation requires an additional quantity friction.

B Appendix Figures and Tables

This appendix presents robustness tests for the baseline results of the paper.

Table A1 relaxes the baseline treatment of currency unions. In the baseline, intra-euro-area trade is excluded from the partner sets Ω^X and Ω^M , since countries sharing a currency have no bilateral exchange-rate variation. Here, euro-area members remain distinct trading partners. Thus, in this specification, each country is treated separately even if they share a common currency. The main findings are unchanged. The coefficient on the export-weighted return remains negative, that on the import-weighted return remains positive, and Δq^{XM} is generally positive and significant. Trade exposure also matters, consistent with the main result.

Table A2 examines whether the bilateral currency effects vary across sectors with different degrees of trade exposure. Panel A splits the sample at the median industry-level trade share into tradable and non-tradable sectors. The panel suggests that the export channel (Δq^X) is negative and statistically significant for tradable industries but is economically small and statistically insignificant for non-tradable industries. The import channel (Δq^M) is positive and significant in both subsamples, although the coefficient is slightly larger for tradable industries. The export-to-import return shows the same pattern, with a positive and statistically significant coefficient for tradable industries but an insignificant one for non-tradable industries. Panel B shows an analogous conclusion when the sample is split between goods-producing and service-sector industries. Overall, these subsample patterns are consistent with a bilateral trade-channel interpretation of the main findings.

Table A3 splits the country-industry sample into more open and less open economies. Countries are classified using their full-sample economy-wide trade openness, defined as the ratio of total exports plus imports to total output plus intermediate inputs. Overall, the results are consistent with bilateral currency effects, but the effect is more evident in more open economies. Export-weighted currency appreciation is more strongly associated

with lower returns, and import-weighted appreciation with higher returns, than in less open economies, where the effects are small and statistically indistinguishable from zero.

Table A4 replicates the country-level regressions from the main text, treating each country as a distinct trading partner regardless of currency union membership. Both the OECD Input-Output Table (Panel A) and DOTS (Panel B) estimates are qualitatively similar to the baseline, suggesting that the main country-level signs are not driven solely by the treatment of intra-currency-union trade flows.

Figure A1 extends the invoicing analysis in Table 5 with local projections of cumulative country-industry revenue and COGS growth through horizons of up to eight years. The figure plots the coefficient on the contemporaneous bilateral currency shock interacted with the relevant USD-invoicing share, while controlling for the lagged shock and its interaction. The interaction is positive at the one-year horizon in both panels but becomes insignificant at later horizons, consistent with USD invoicing delaying pass-through rather than providing a permanent hedge of currency movements.

Table A1: Country-industry robustness
Euro-area countries as distinct trading partners

This table replicates Table 2 using an alternative construction of the trade-weighted currency returns in which each country is treated as a distinct trading partner regardless of currency union membership. Columns (1)–(4) report value-weighted returns and columns (5)–(8) report equally weighted returns. Panel A presents results using export/import-weighted returns, and Panel B uses export-to-import currency returns ($\Delta q_{c,i,t}^{XM}$) and the USD return ($\Delta q_{c,t}$) as defined in the main text. Odd-numbered columns include country $\times$ industry fixed effects; even-numbered columns add country $\times$ year fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. Using export/import-weighted returns

	Value-weighted returns				Equally weighted returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^X$	-0.224 (-3.17)	-0.199 (-3.14)	-0.152 (-2.42)	-0.136 (-2.37)	-0.229 (-3.12)	-0.203 (-3.14)	-0.161 (-2.44)	-0.144 (-2.39)
$\Delta q_{c,i,t}^M$	0.297 (3.93)	0.273 (4.09)	0.210 (2.97)	0.194 (3.15)	0.328 (4.42)	0.287 (4.53)	0.246 (3.44)	0.213 (3.48)
$R_{c,t}$			0.614 (14.97)	0.608 (15.37)			0.575 (15.13)	0.565 (15.26)
$R_{WLD,t}$	0.812 (14.99)	0.786 (16.14)	0.282 (6.51)	0.270 (6.60)	0.770 (13.21)	0.738 (13.97)	0.274 (5.85)	0.258 (5.74)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.108	0.130	0.166	0.182	0.107	0.134	0.161	0.183
N	299,688	299,688	299,688	299,688	299,688	299,688	299,688	299,688

Panel B. Using export-to-import currency returns

	Value-weighted returns				Equally weighted returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^{XM}$	0.257 (3.63)	0.230 (3.72)	0.181 (2.77)	0.163 (2.82)	0.276 (3.86)	0.240 (3.94)	0.205 (3.04)	0.178 (3.00)
$\Delta q_{c,t}$	0.015 (0.33)	0.005 (0.12)	0.053 (1.47)	0.043 (1.56)	0.047 (1.08)	0.025 (0.67)	0.083 (2.19)	0.060 (2.03)
$R_{c,t}$			0.615 (14.89)	0.609 (15.26)			0.577 (15.03)	0.567 (15.14)
$R_{WLD,t}$	0.812 (14.76)	0.789 (15.88)	0.271 (5.96)	0.261 (6.10)	0.763 (12.85)	0.736 (13.48)	0.256 (5.16)	0.245 (5.11)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.108	0.130	0.166	0.182	0.106	0.134	0.162	0.183
N	299,688	299,688	299,688	299,688	299,688	299,688	299,688	299,688

Table A2: Tradable vs. Non-tradable and Goods vs. Services sector decomposition

This table presents the baseline country-industry stock return regressions from Table 2 estimated separately for the tradable and non-tradable sectors (Panel A), and the goods and services sectors (Panel B). Industry tradability is determined by comparing each country-industry's time-averaged trade share with the median of the distinct country-industry averages within the same country; this classification is time-invariant. The goods sector consists of industries corresponding to primary goods and manufacturing, while the remaining industries comprise the services sector. Odd columns report the estimates using export- and import-weighted currency returns (Δq^X and Δq^M), and even columns use the export-to-import currency return (Δq^{XM}). All specifications include country $\times$ industry fixed effects. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. Tradable and non-tradable sectors

	Value-weighted returns				Equally weighted returns			
	Tradable		Non-tradable		Tradable		Non-tradable	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^X$	-0.134		-0.035		-0.100		-0.035	
	(-2.85)		(-0.99)		(-2.24)		(-0.97)	
$\Delta q_{c,i,t}^M$	0.144		0.099		0.176		0.107	
	(2.64)		(2.44)		(3.57)		(2.76)	
$\Delta q_{c,i,t}^{XM}$		0.133		0.046		0.118		0.047
		(2.83)		(1.40)		(2.49)		(1.36)
$\Delta q_{c,t}$		-0.032		0.010		0.036		0.023
		(-0.88)		(0.32)		(0.99)		(0.68)
$R_{WLD,t}$	0.858	0.867	0.756	0.757	0.812	0.808	0.731	0.730
	(15.59)	(15.49)	(14.26)	(14.27)	(13.74)	(13.45)	(12.35)	(12.25)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.145	0.145	0.112	0.112	0.130	0.130	0.104	0.104
N	155,541	155,541	158,915	158,915	155,541	155,541	158,915	158,915

Panel B. Goods and services sectors

	Value-weighted returns				Equally weighted returns			
	Goods sector		Services sector		Goods sector		Services sector	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^X$	-0.159		0.010		-0.125		0.006	
	(-3.13)		(0.17)		(-2.54)		(0.12)	
$\Delta q_{c,i,t}^M$	0.149		0.087		0.176		0.097	
	(2.54)		(1.52)		(3.25)		(1.87)	
$\Delta q_{c,i,t}^{XM}$		0.155		0.006		0.138		0.009
		(3.09)		(0.11)		(2.64)		(0.20)
$\Delta q_{c,t}$		-0.027		0.010		0.033		0.024
		(-0.75)		(0.31)		(0.89)		(0.72)
$R_{WLD,t}$	0.829	0.835	0.782	0.785	0.787	0.782	0.754	0.754
	(13.77)	(13.77)	(16.14)	(15.97)	(12.39)	(12.21)	(13.69)	(13.46)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.130	0.130	0.126	0.126	0.118	0.118	0.115	0.115
N	168,658	168,658	145,798	145,798	168,658	168,658	145,798	145,798

Table A3: Country-industry stock returns by country trade openness

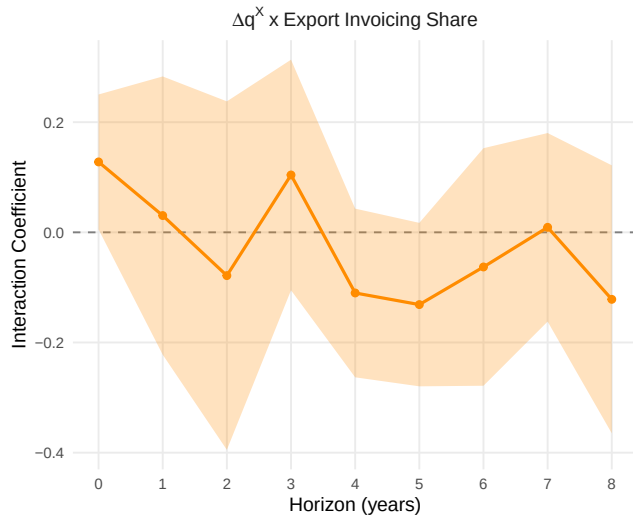
This table replicates the baseline country-industry return regressions in Table 2, separately for more open and less open economies. A country's trade openness is the ratio of its total exports plus imports to its total output plus intermediate inputs, each summed to the country level. Countries above the median are classified as more open. Odd columns use export- and import-weighted currency returns (Δq^X and Δq^M), while even columns use the export-to-import currency return (Δq^{XM}) and the USD return (Δq). All specifications include country $\times$ industry fixed effects and the world MSCI return. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

	Value-weighted returns				Equally weighted returns			
	More open economies		Less open economies		More open economies		Less open economies	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,i,t}^X$	-0.420		0.013		-0.383		0.024	
	(-3.83)		(0.27)		(-3.46)		(0.56)	
$\Delta q_{c,i,t}^M$	0.563		-0.061		0.560		-0.034	
	(5.37)		(-1.04)		(5.48)		(-0.64)	
$\Delta q_{c,i,t}^{XM}$		0.478		-0.017		0.456		-0.025
		(4.54)		(-0.35)		(4.35)		(-0.58)
$\Delta q_{c,t}$		0.071		-0.070		0.090		-0.015
		(1.51)		(-1.77)		(1.83)		(-0.41)
$R_{WLD,t}$	0.806	0.799	0.801	0.815	0.776	0.765	0.761	0.764
	(13.20)	(12.69)	(16.11)	(16.62)	(11.71)	(11.32)	(13.91)	(13.94)
Country $\times$ Industry FE	Y	Y	Y	Y	Y	Y	Y	Y
R ²	0.121	0.121	0.137	0.138	0.113	0.113	0.122	0.122
N	153,291	153,291	161,165	161,165	153,291	153,291	161,165	161,165

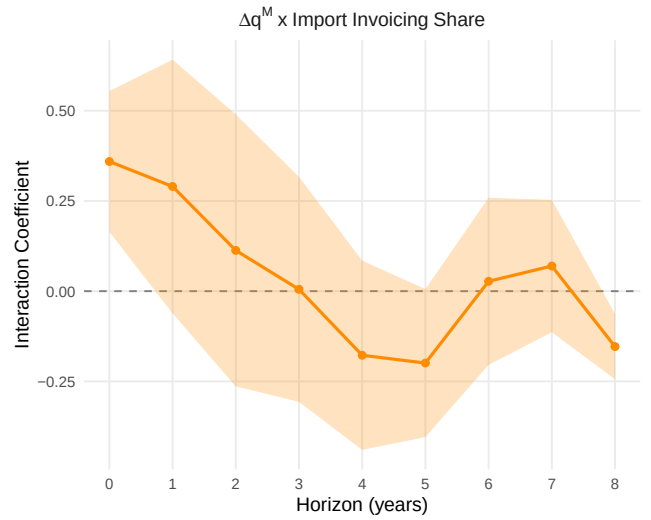
Table A4: Country-level robustness
Euro-area countries as distinct trading partners

This table replicates Table 6 treating each country as a distinct trading partner regardless of currency union membership. Columns (1)–(2) and (5)–(6) use the export- and import-weighted currency returns $\Delta q_{c,t}^X$ and $\Delta q_{c,t}^M$ separately, and columns (3)–(4) and (7)–(8) report the export-to-import differential $\Delta q_{c,t}^{XM}$ alongside the bilateral USD return $\Delta q_{c,t}$. The dependent variable in columns (1)–(4) is the raw country index return, and in columns (5)–(8) it is the return in excess of the US equity return. Odd-numbered columns include country fixed effects; even-numbered columns add country $\times$ year fixed effects. Panel A uses OECD Input-Output trade weights and Panel B uses IMF DOTS trade flows. T-statistics in parentheses use Driscoll–Kraay standard errors with a lag of 3.

Panel A. OECD Input-Output sample								
	Country index returns				Country excess-to-US returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}^X$	-0.964 (-3.26)	-0.942 (-3.33)			-0.787 (-2.80)	-0.778 (-3.01)		
$\Delta q_{c,t}^M$	1.010 (3.37)	0.995 (3.43)			0.821 (2.88)	0.813 (3.08)		
$\Delta q_{c,t}^{XM}$			0.928 (3.12)	0.913 (3.20)			0.807 (2.69)	0.780 (2.88)
$\Delta q_{c,t}$			-0.032 (-0.90)	-0.028 (-0.71)			0.011 (0.27)	-0.001 (-0.02)
$R_{WLD,t}$	0.870 (24.16)	0.856 (26.35)	0.880 (23.34)	0.866 (24.37)				
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.413	0.469	0.413	0.469	0.008	0.105	0.008	0.105
N	7,653	7,653	7,653	7,653	7,327	7,327	7,327	7,327
Panel B. IMF DOTS sample								
	Country index returns				Country excess-to-US returns			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\Delta q_{c,t}^X$	-1.333 (-5.01)	-1.241 (-4.56)			-0.995 (-3.41)	-0.896 (-2.94)		
$\Delta q_{c,t}^M$	1.397 (5.14)	1.307 (4.69)			1.043 (3.52)	0.939 (3.05)		
$\Delta q_{c,t}^{XM}$			1.268 (4.50)	1.177 (4.12)			0.997 (3.20)	0.869 (2.76)
$\Delta q_{c,t}$			-0.029 (-0.78)	-0.027 (-0.68)			0.001 (0.03)	-0.011 (-0.29)
$R_{WLD,t}$	0.853 (25.73)	0.842 (27.89)	0.866 (24.70)	0.855 (25.64)				
Country FE	Y	Y	Y	Y	Y	Y	Y	Y
Country $\times$ Year FE	N	Y	N	Y	N	Y	N	Y
R ²	0.413	0.469	0.412	0.469	0.011	0.106	0.010	0.106
N	6,675	6,675	6,675	6,675	6,349	6,349	6,349	6,349



(a) Revenue: export-weighted shock and export invoicing



(b) COGS: import-weighted shock and import invoicing

Figure A1: USD invoicing and the dynamic response of revenue and COGS to bilateral currency shocks

This figure illustrates cumulative revenue and COGS growth from $t - 1$ through $t + h$, for $h = 0, \dots, 8$, estimated by local projections. Panels (a) and (b) plot the coefficients on the contemporaneous export- and import-weighted currency returns, respectively, interacted with the corresponding USD-invoicing shares. All regressions include country-industry fixed effects, the lagged currency shock and its invoicing interaction, current and lagged USD currency returns, the lagged outcome, and current and lagged global outcome controls. Shaded bands are 90% confidence intervals based on Driscoll–Kraay standard errors with three lags.